Stellar activity cycles from ground- and space-based observations

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KONKOLY

HUN-REN
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Motivation

- dynamo observables are crucial in model parametrization
- serving reliable observational input constraints for stellar/solar dynamo theories
- solar-stellar similarities and differences
- better understanding of stellar structure and evolution

Stellar dynamo observables:

- long-term activity cycles
- active longitudes (e.g. “flip-flop” with 2 permanent active longitudes, Elstner & Korhonen 2005)
- surface differential rotation (Doppler maps, Fourier-spectra)
- meridional flow (? Doppler-maps)

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- Schwabe (1844) – there is a periodicity in sunspot numbers
- Wolf (1852) – determines the length of the period to be 11.1 years

Schon aus meinen früheren Beobachtungen, die ich jährlich in dieser Zeitschrift mittheile, scheint sich eine gewisse Periodicität der Sonnenflecken zu ergeben und diese Wahrscheinlichkeit gewinnt durch die diesjährigen noch an Sicherheit. Obgleich ich schon in Band 15. Nr. 350 pag. 246 der Astr. Nachrichten die Menge der Gruppen in den Jahren 1826 bis 1837 angab, so füge ich doch hier noch ein vollständiges Verzeichnifs aller meiner bisher beobachteten Sonnenflecken bei, worin ich neben der Zahl der Gruppen auch die Zahl der Beobachtungstage und der fleckenfreien Tage angemerkt habe. Die Zahl der Gruppen allein giebt nämlich keine hinreichende Genauigkeit zur Beurtheilung einer Periode, weil ich mich überzeugt habe, dafs bei sehr starken Anhäufungen der Sonnenflecken eine etwas zu geringe bei dem sparsamen Erscheinen derselben eine etwas zu grofse Anzahl der Gruppen gerechnet wird. Im ersten Falle fliefsen oft mehrere Gruppen zu einer einzigen zusammen und im zweiten trennt sich leicht eine Gruppe, durch Auflösung einiger Flecken, in zwei einzelne. Hiermit wird wohl die Wiederholung des frühern Verzeichnisses entschuldigt sein.

Jahr.	Gruppen.	Fleckenfreie tage.	Beobachtungstage.
1826	118	22	277
1827	161	2	273
1828	225	0	282

Schwabe, 1844

Ich sende Ihnen hier für die Astronom. Nachrichten einen kurzen Bericht über eine grössere, sehr zeitraubende Arbeit, die ich nächstens unserer Naturforschenden Gesellschaft vorlegen, und dann wahrscheinlich später in die Mittheilungen einrücken werde. Es sind neue Untersuchungen über die Periode der Sonnenflecken und ihre Bedeutung, gestützt auf alle Sonnenflecken-Beobachtungen älterer und neuerer Zeit, die ich aus wenigstens 400 Bänden astronomischer Werke, Abhandlungen und Zeitschriften zusammensuchte. Die Arbeit zerfällt in 6 Abschnitte:

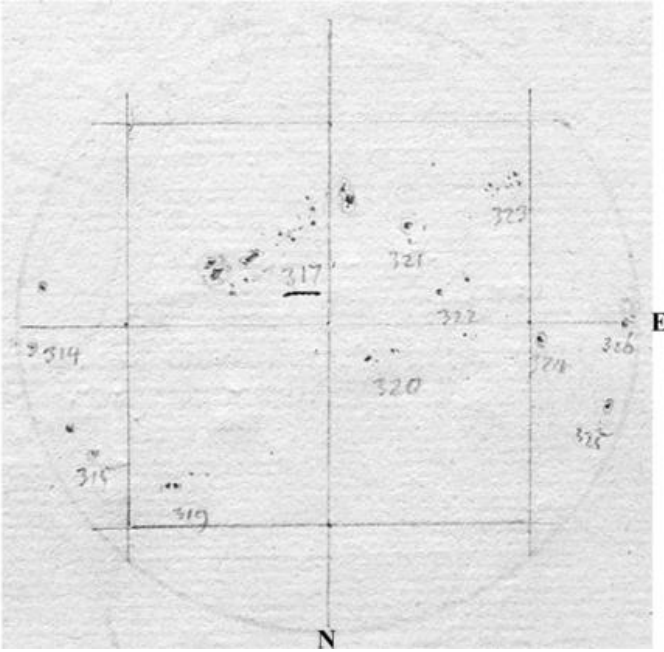
Im ersten Abschnitte setze ich 16 Epochen für das Maximum oder Minimum der Sonnenflecken aus diesen Beobachtungen fest, und ermittle daraus die Länge der Sonnenfleckenperiode zu

$$11{,}111 \pm 0{,}038 \text{ Jahren.}$$

(...)

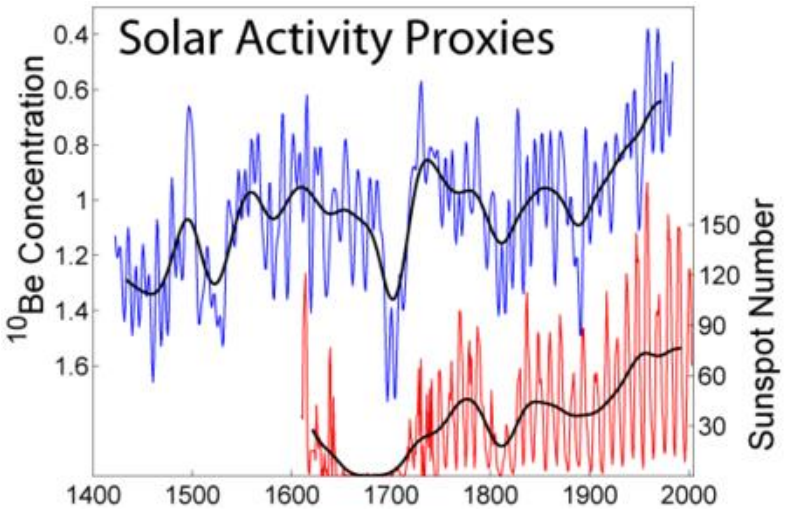
Bern 1852, Nov. 2.

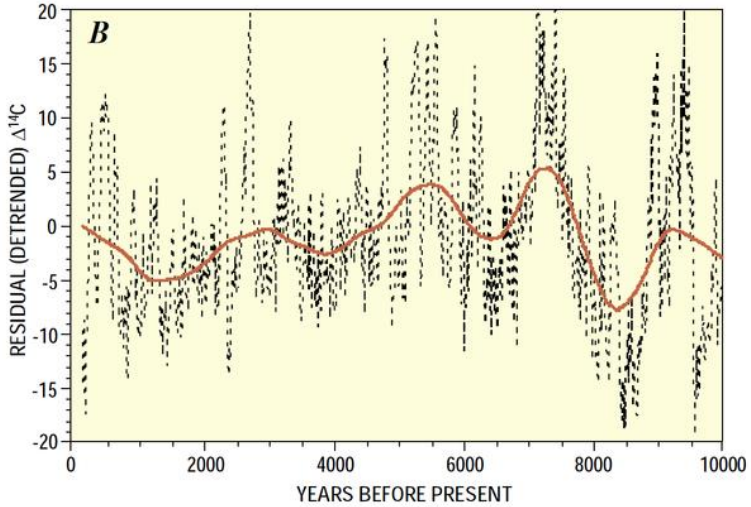
Rudolf Wolf.



Longer solar cycles

- 11/22 years – Schwabe/Hale cycle
- 70-100 years – Gleissberg-cycle (^{14}C, ^{10}Be isotopes in ice sheets/tree rings, Usoskin+ 2012)
- ~200 years – Suess-cycle (radiocarbon proxies)
- ~2400 years – Hallstatt-cycle(?)
- ~6000 years?

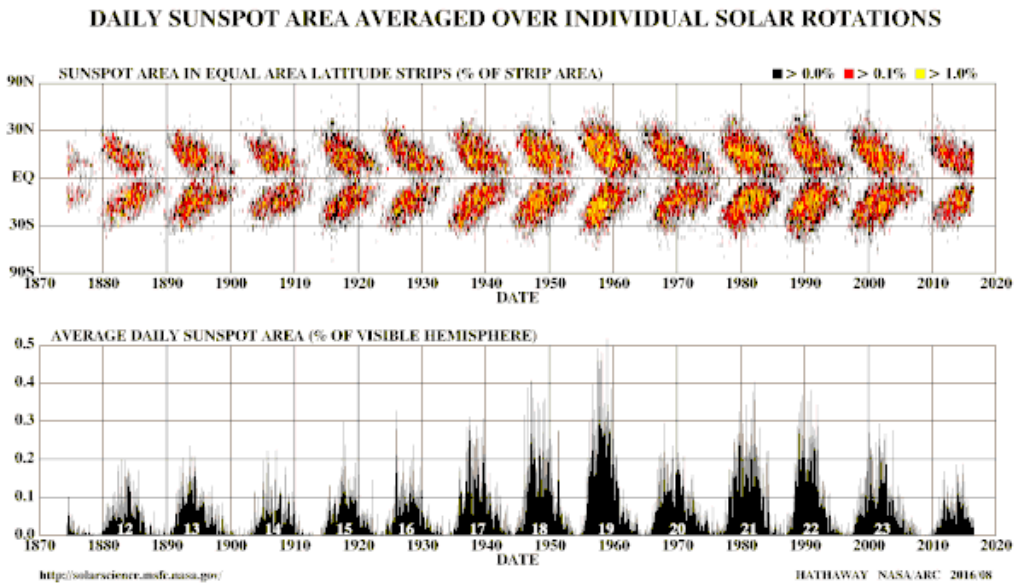


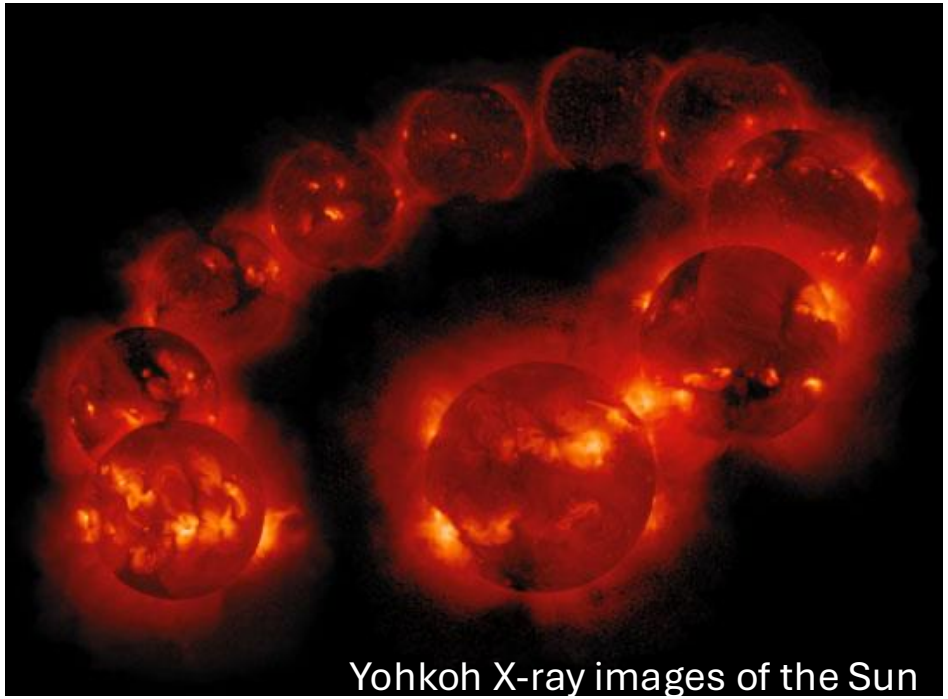


Hallstatt cycle from ^{14}C measurements

Tracers of the solar cycle

- Sunspot activity – spot number, average latitude of spots, total sunspot area
- Chromospheric activity – size/number of chromospheric active regions, emission lines (Ca II H&K)
- Flares (flares and intensity)
- Intensity of X-ray emission
- Structure of the solar corona
- 10.7 cm radio flux
- Magnetic field and solar wind
- Terrestrial tracers




Yohkoh X-ray images of the Sun

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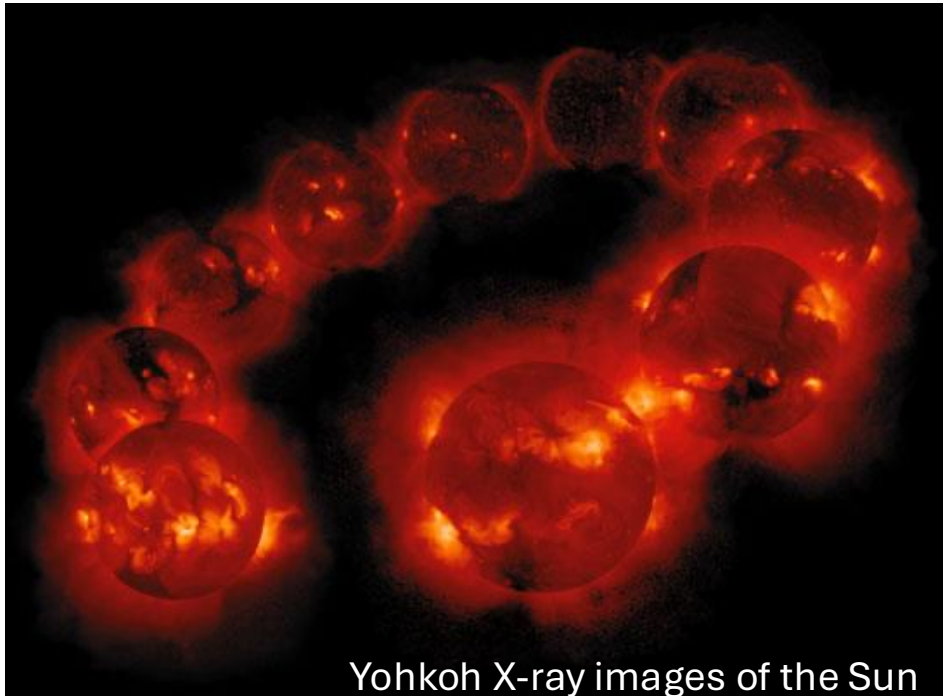
Inconclusive in the literature



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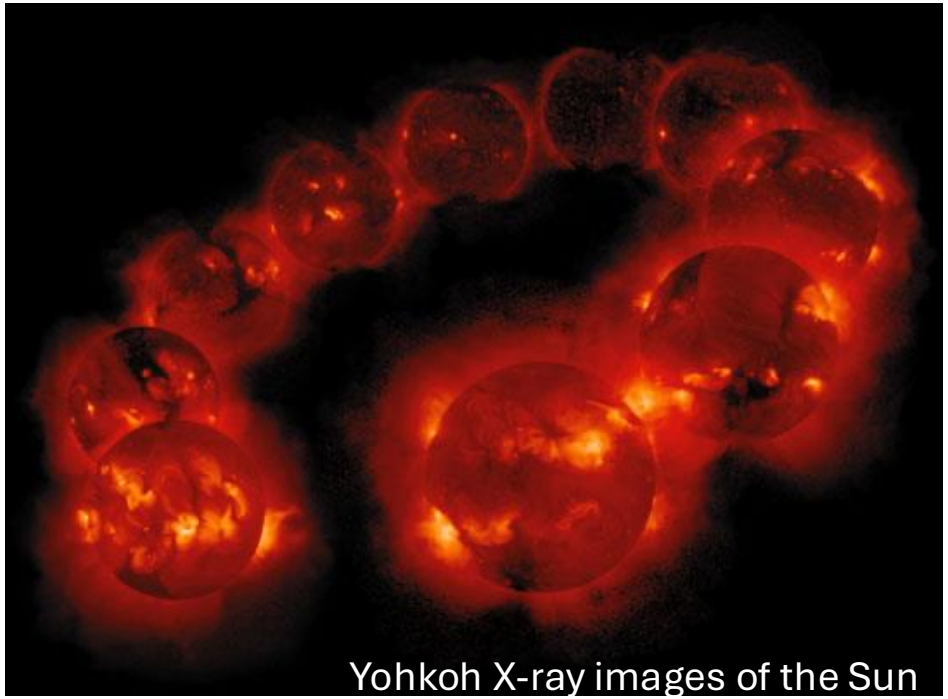


Yohkoh X-ray images of the Sun

Activity cycles on other stars

- Systematic observations are needed – for a long time
- Time-critical phenomenon – better instrumentation won’t help
- Hard to convince funders/institutions to keep projects running for several decades (results won’t come on PhD-timescales)

Cycles on other stars – chromospheric activity

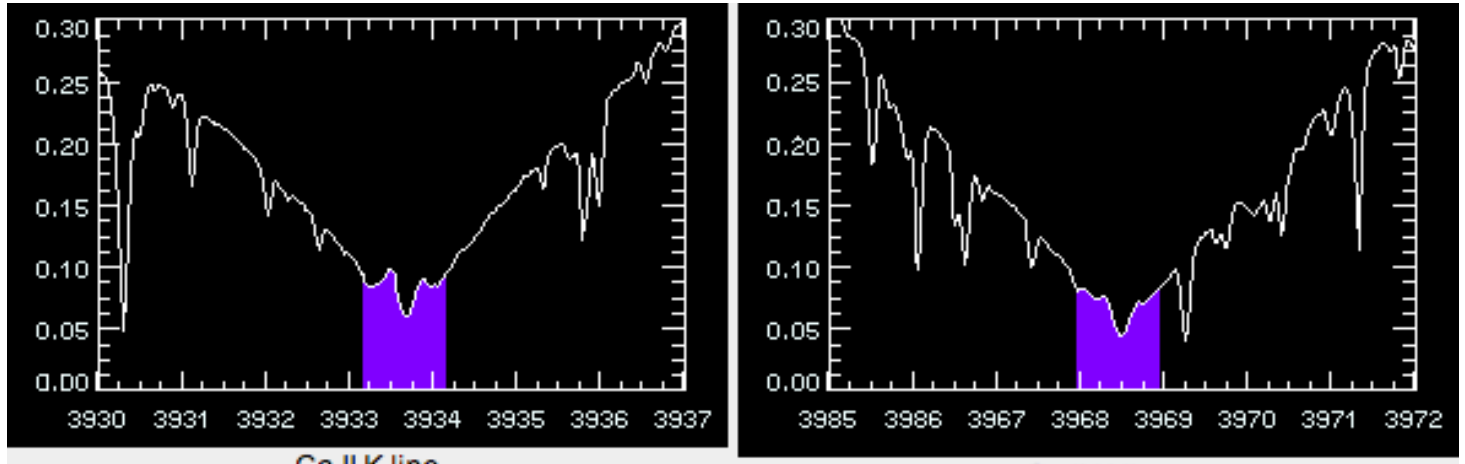
Mt. Wilson, HK Project (1966-2002) – stellar Ca II H&K line variations over decade using the S-index:

$$S = \alpha \frac{H + K}{V + R}$$

Ca II H&K flux

Violet & red continuum

Constant for calibration



Ca II K line

Ca II H line

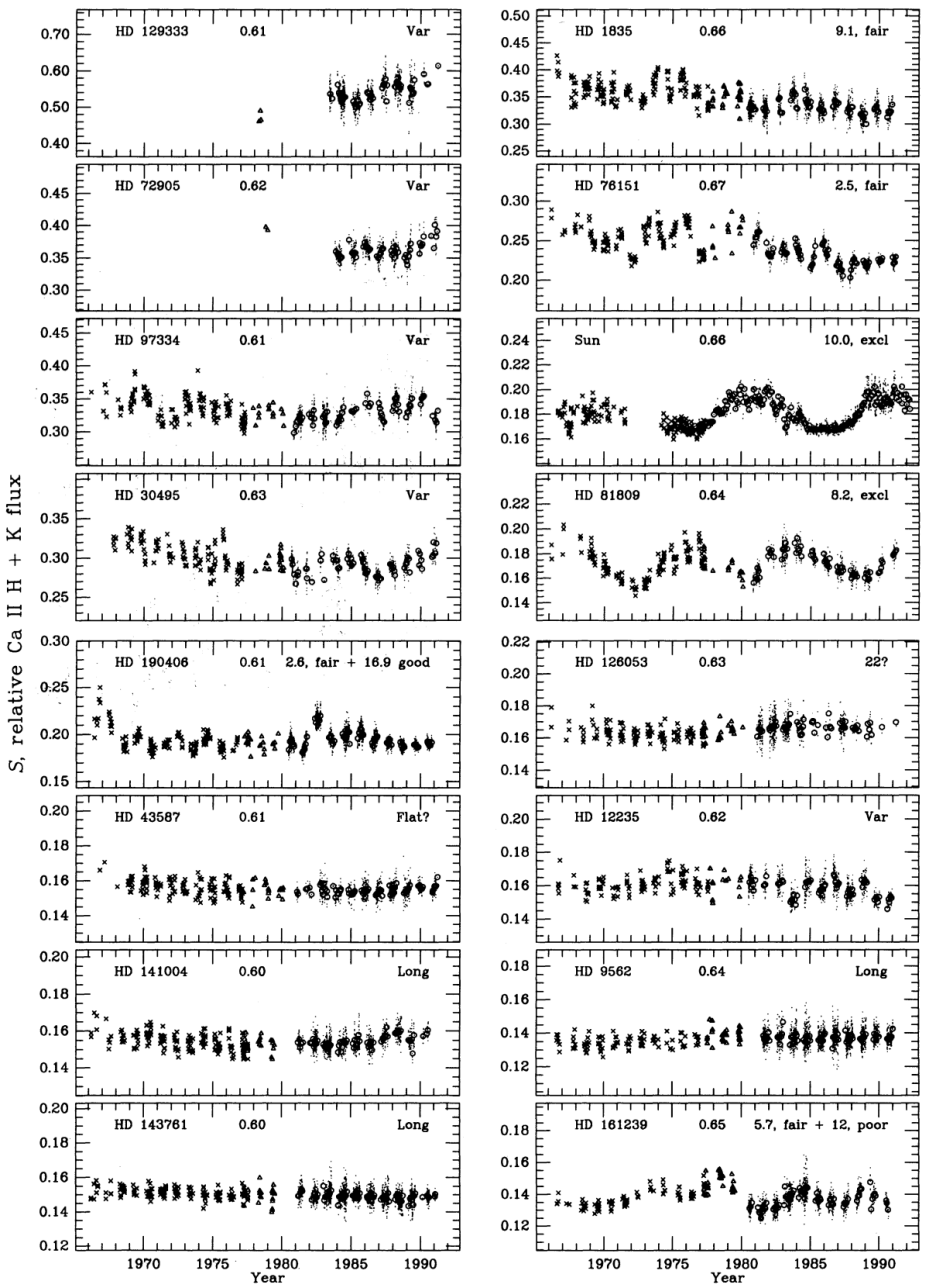


FIG. 1d

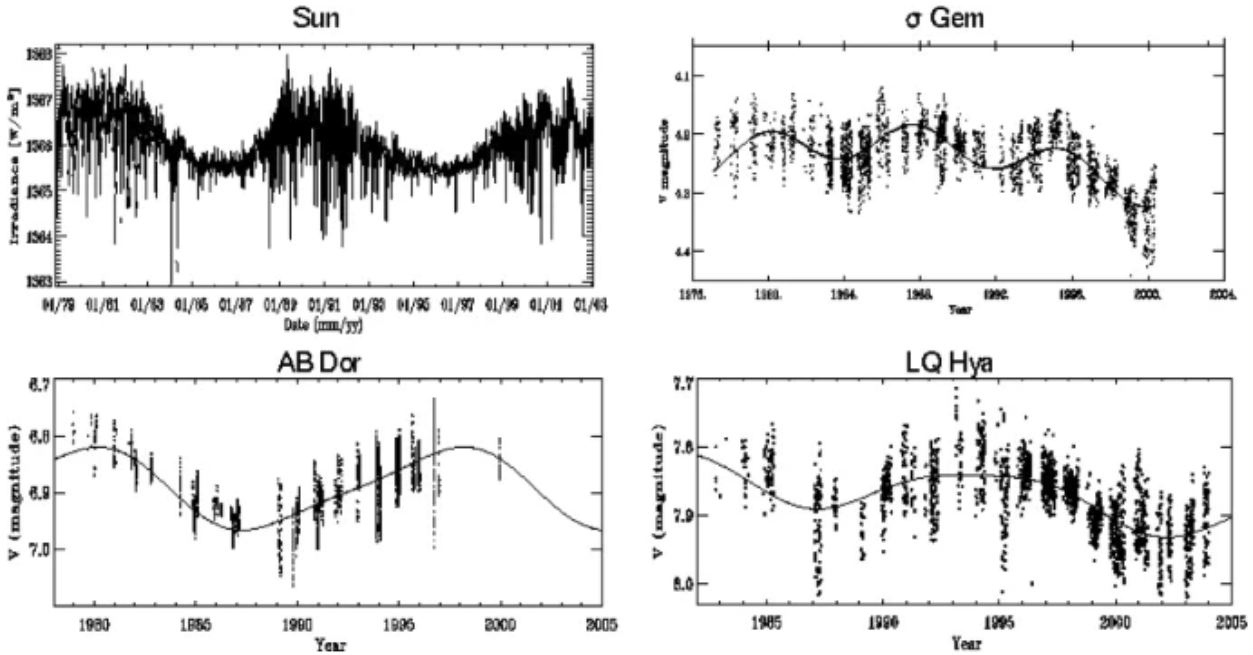
S-index over time on different stars:

- Activity decreases with age and slower rotation,
- older stars have more stable cycles
- Cycle length typically ~10 years (also limited by observations)
- $P_{cyc} \sim Ro^{2.0}$ (Ossendrijver+ 1997)

$$Ro = \frac{P_{rot}}{\tau}$$

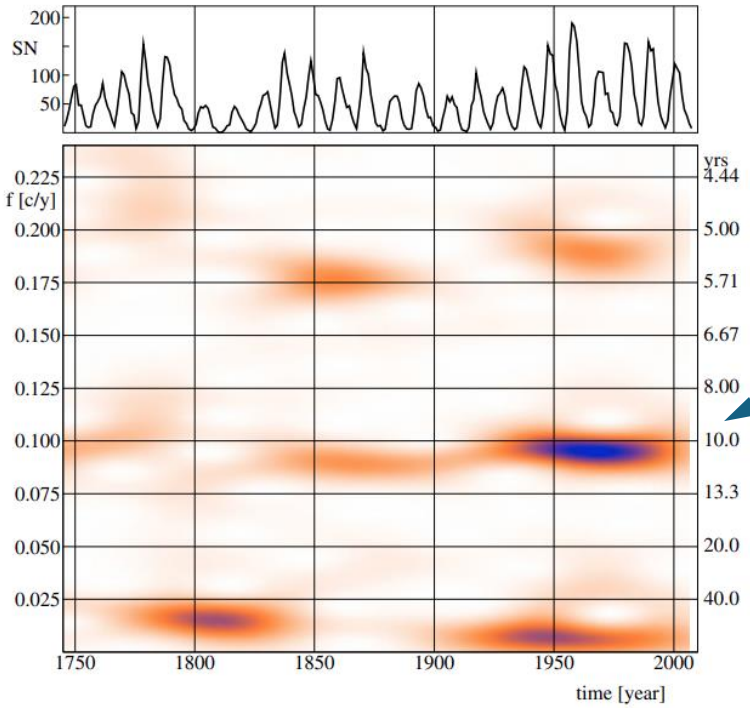
Baliunas et al. (1995)

Cycles on other stars – total spot area (i.e., light curve)

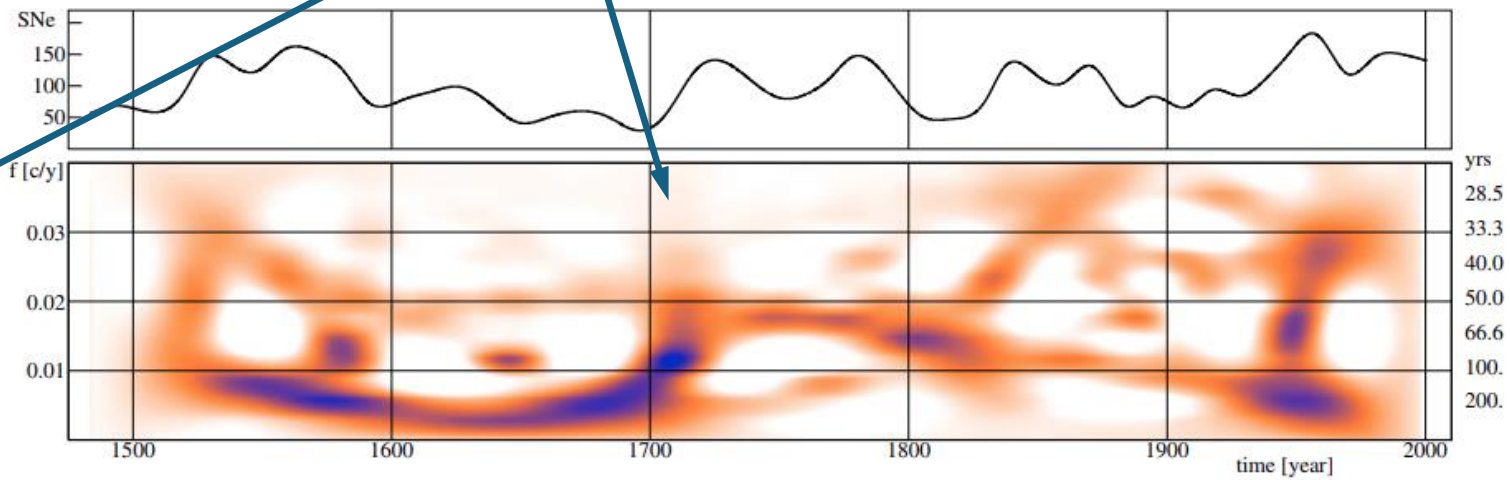


Spot cycles in the solar irradiance and V magnitudes of the RS CVn binary σ Gem and two young solar analogues AB Dor and LQ Hya. Note that the maximum of the spot area corresponds to the maximum irradiance on the Sun and minimum brightness on the stars.

Short-term Fourier transform (STFT)



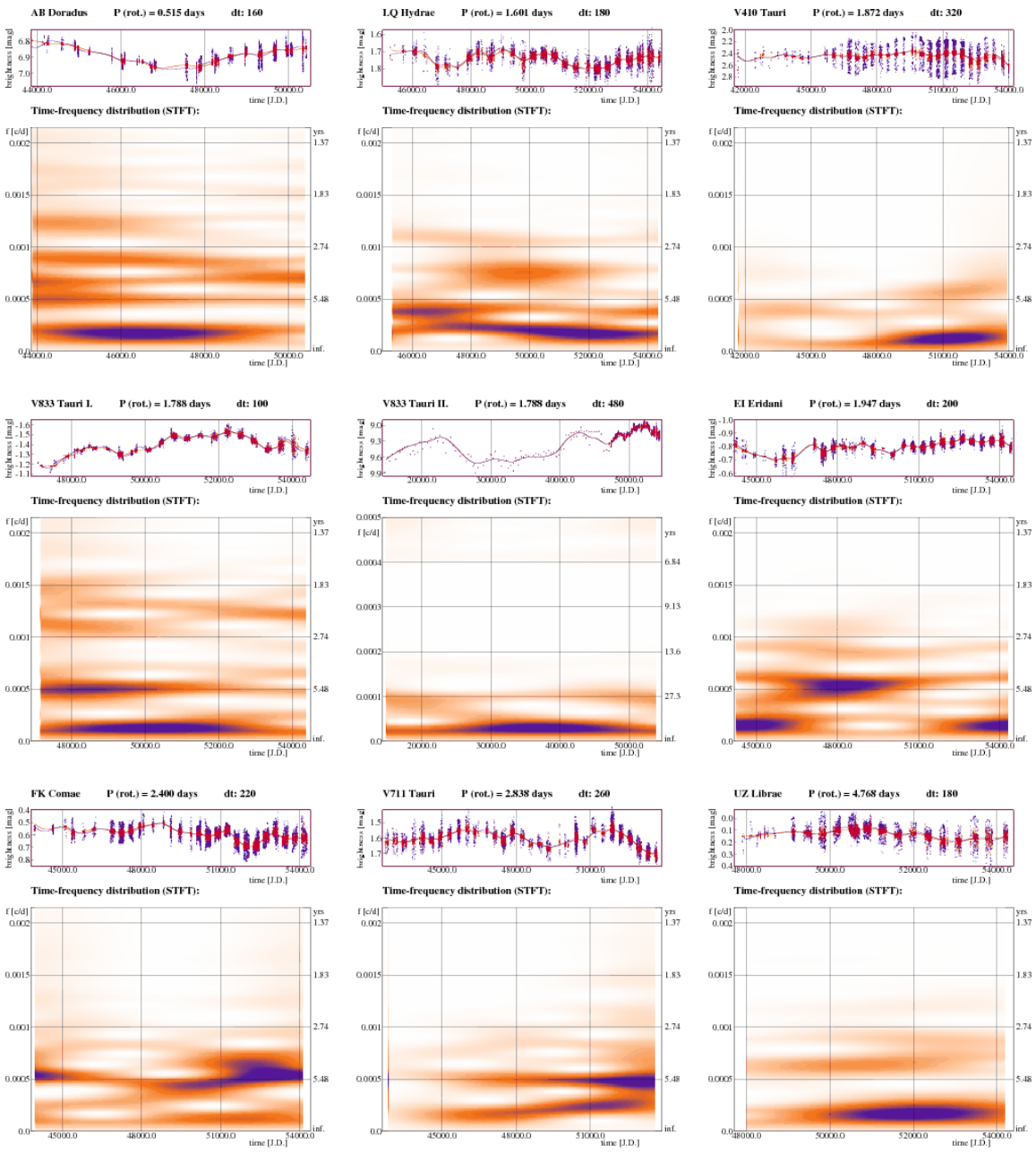
Cycle lengths are changing over time



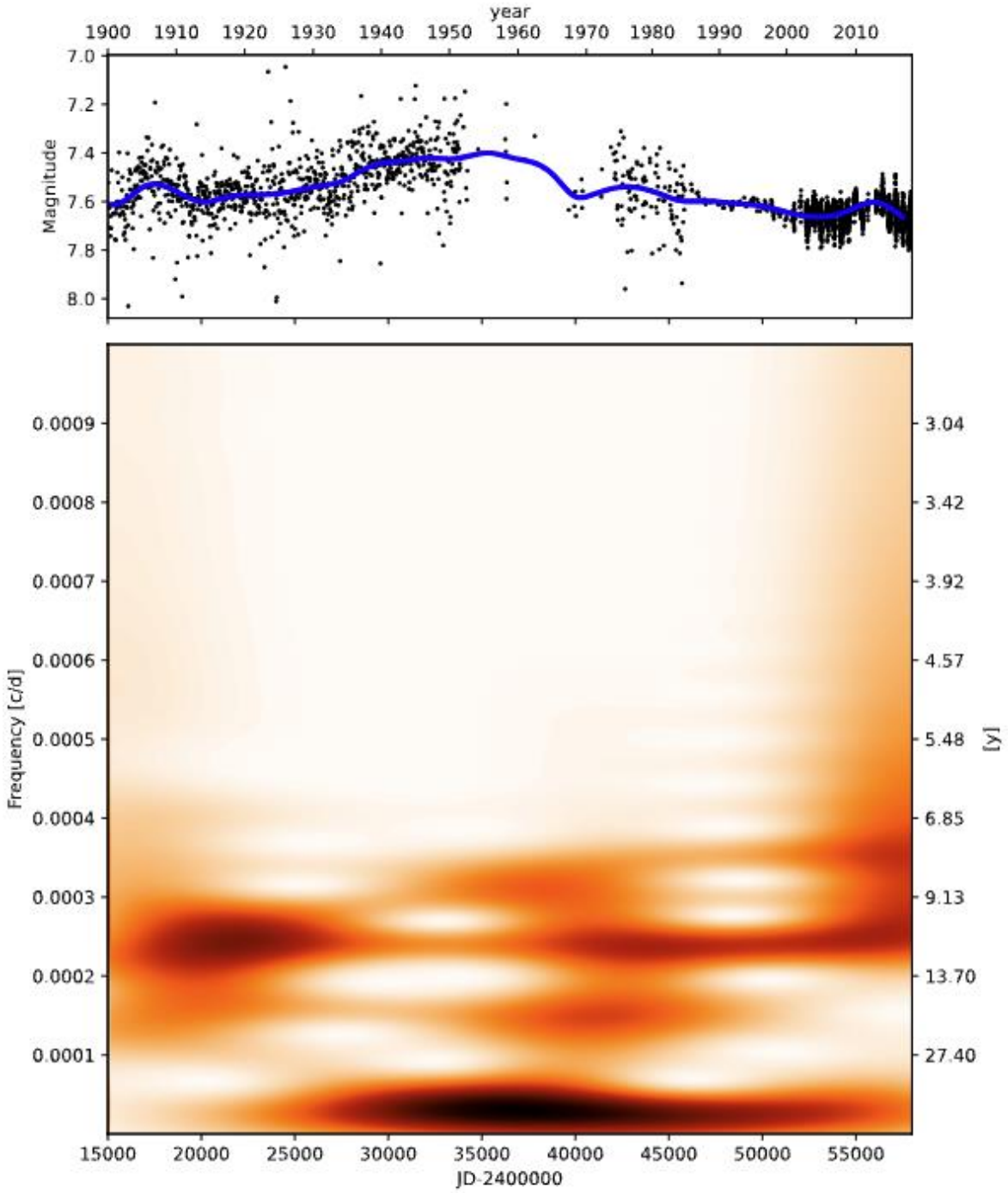
example on sunspot numbers

long-term solar variability during he last 500 years from proxy data

From dwarfs to giants –
multiple and changing cycles are found

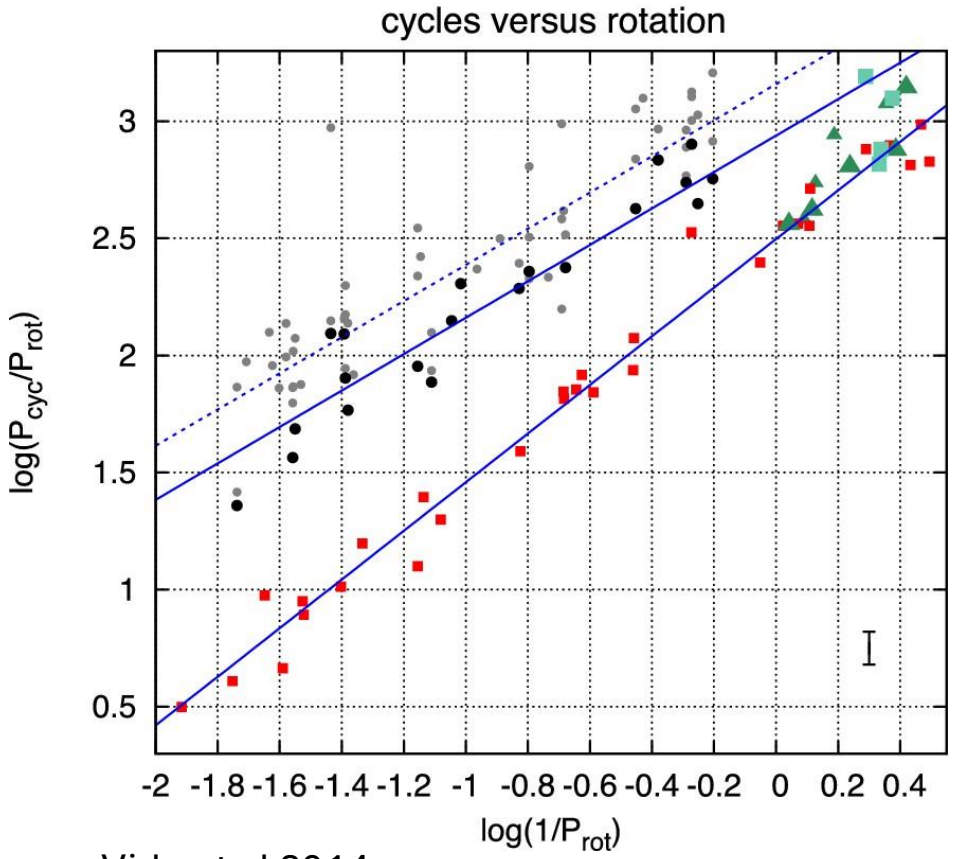


EK Dra – “fast-rotating Sun”
DASCH (scanned Harvard photoplates) &
APT photometry (Messina&Guinan 2012; Järvinen+ 2018)



11-12yr cycle can be seen back to early 1900s!

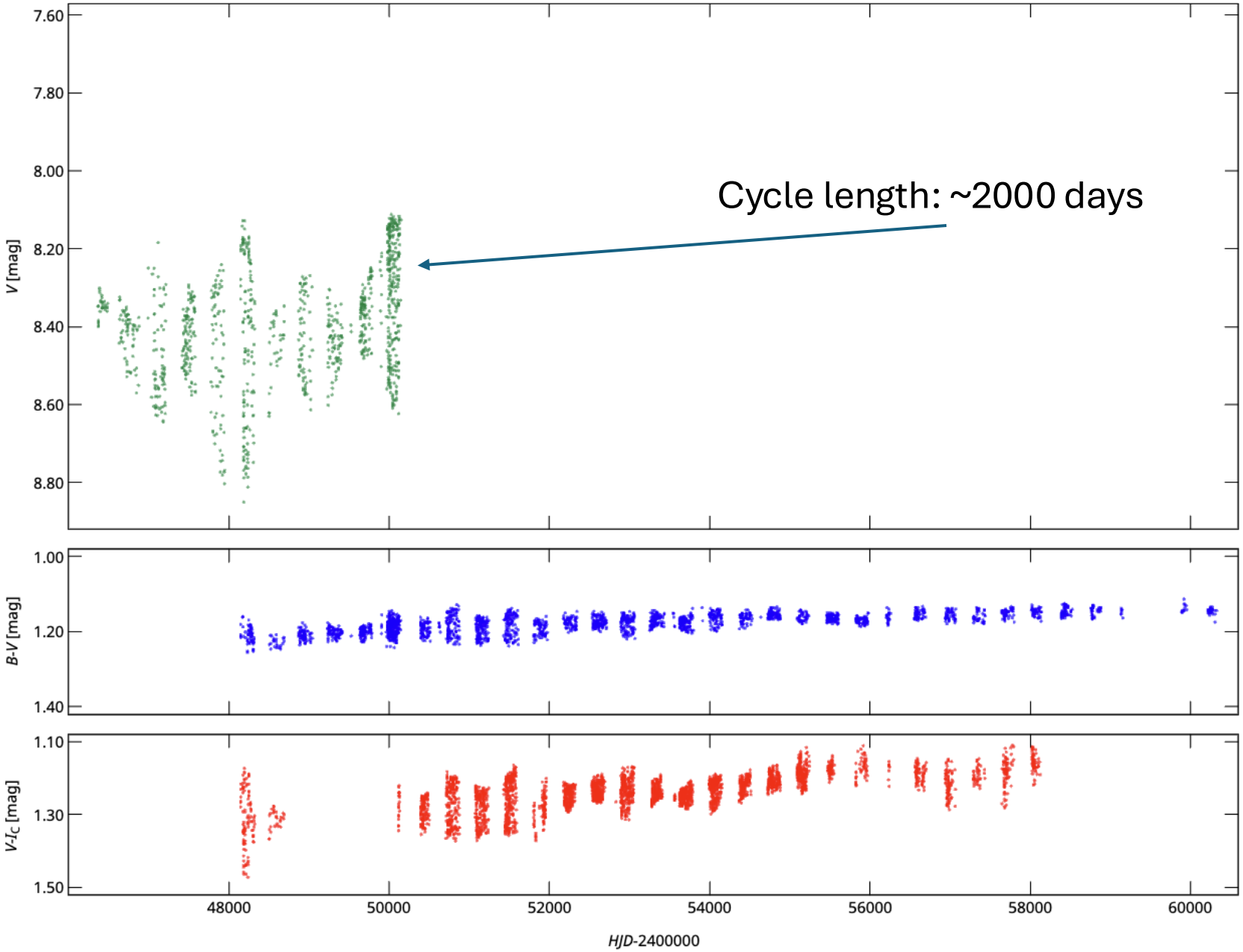
Görgei+ in prep



Vida et al 2014

Generally, fast-rotating stars are expected to have shorter cycles (but that might not be true for every star! Maybe properties of the convection zone is more important?)

We should be careful when interpreting data – XX Tri (K0III SB)



Kővári et al. 2025, submitted

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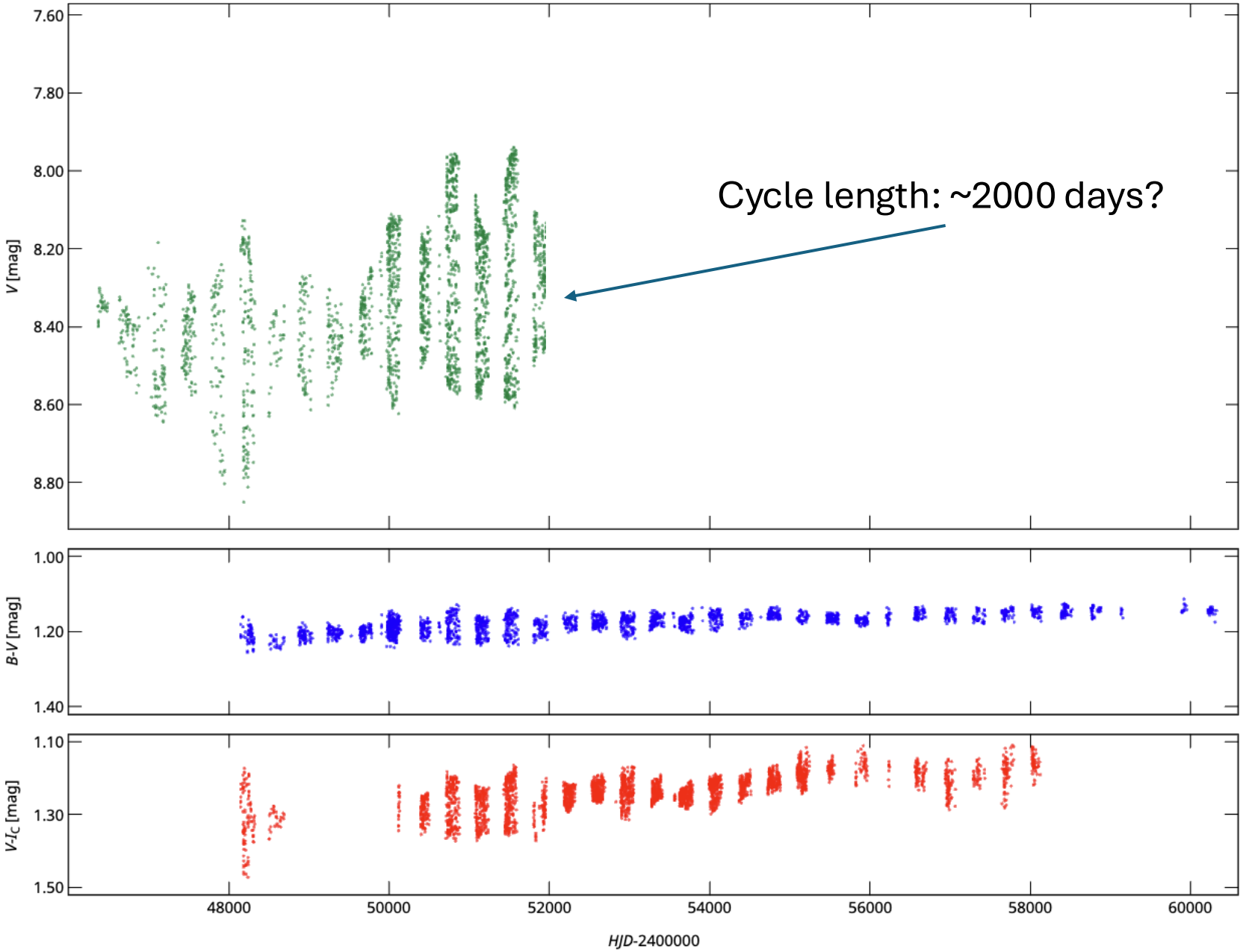

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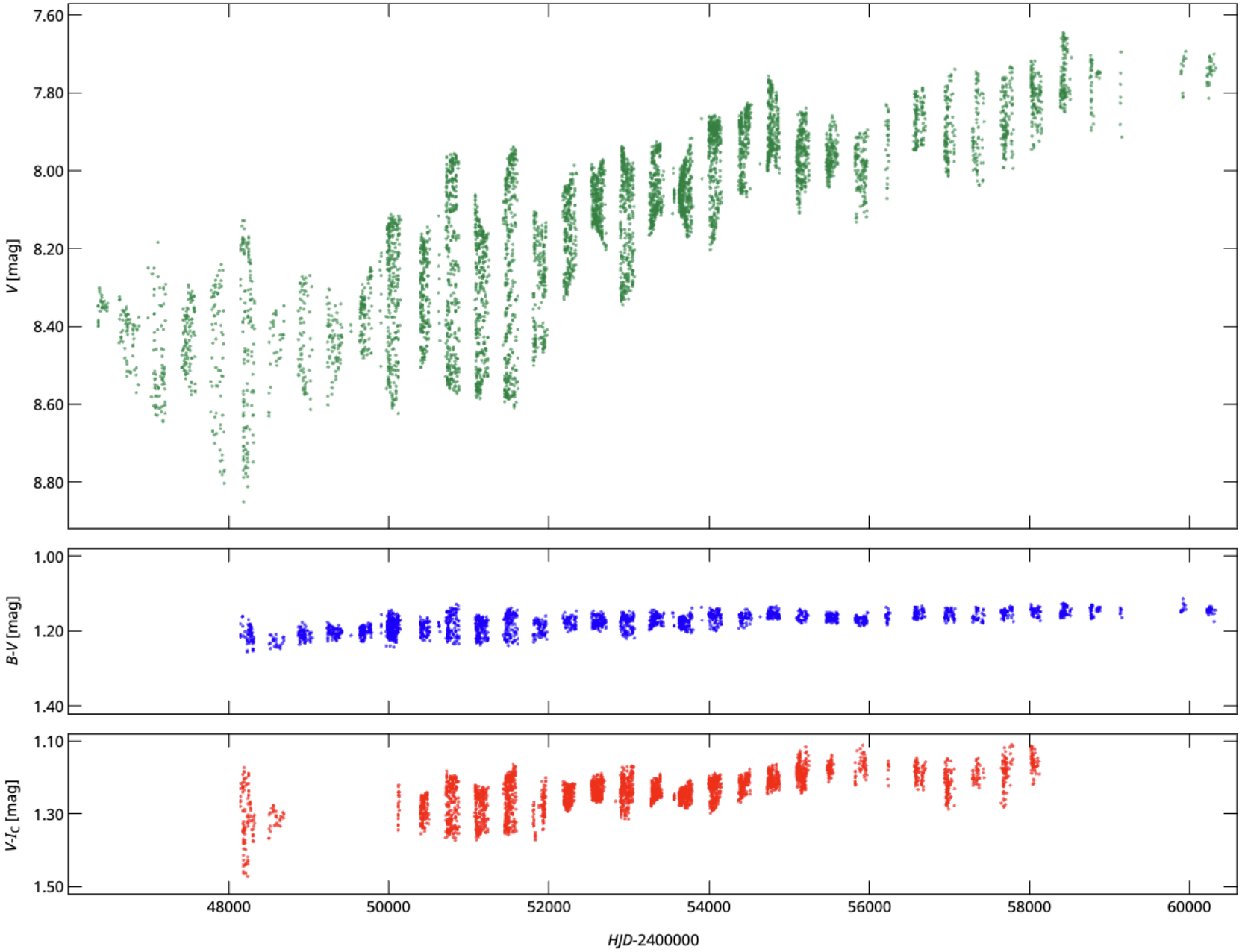
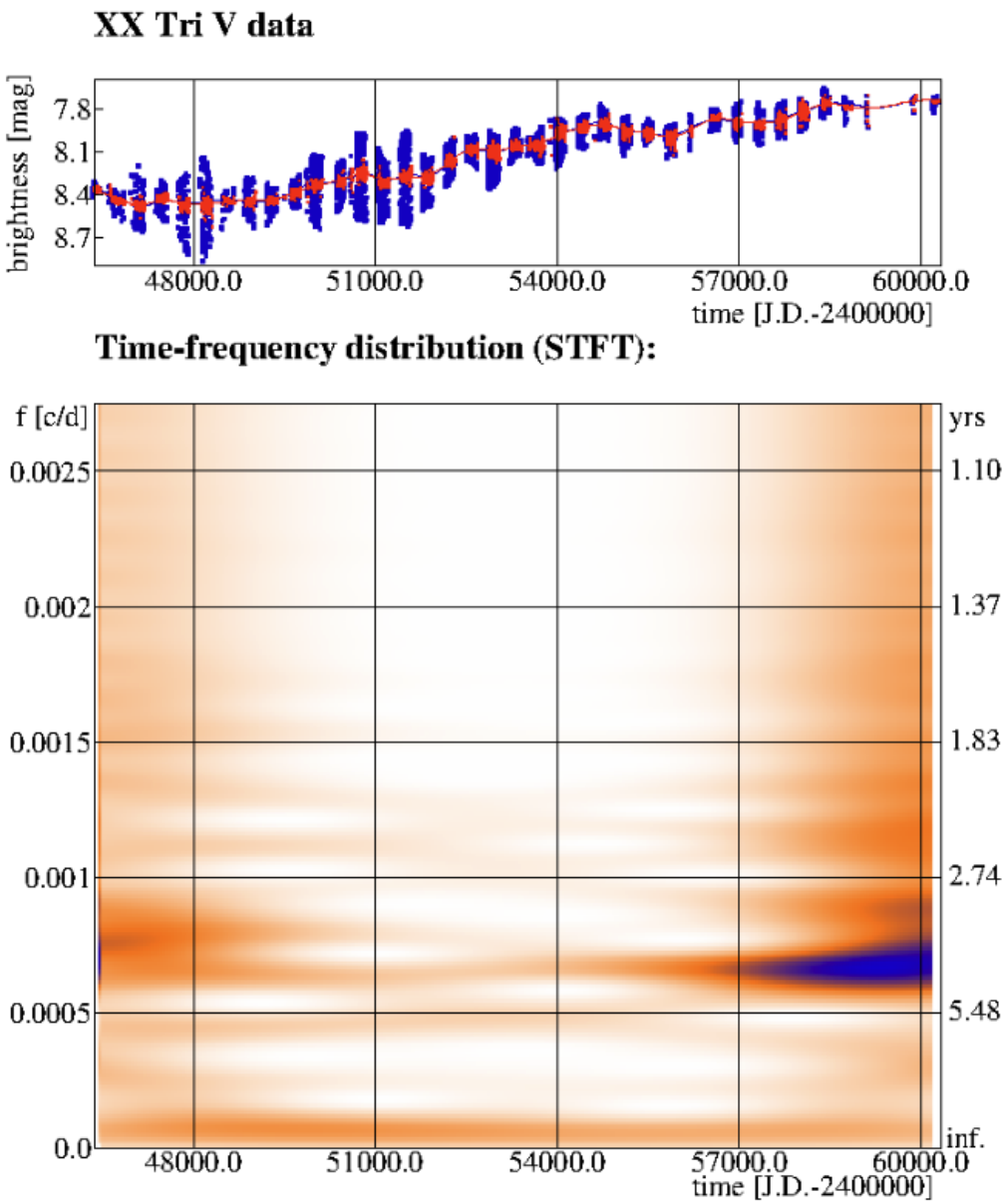


Kővári et al. 2025, submitted

There is a secular trend
+ a decade-long modulation
+ a ~4 year-long cycle with changing length



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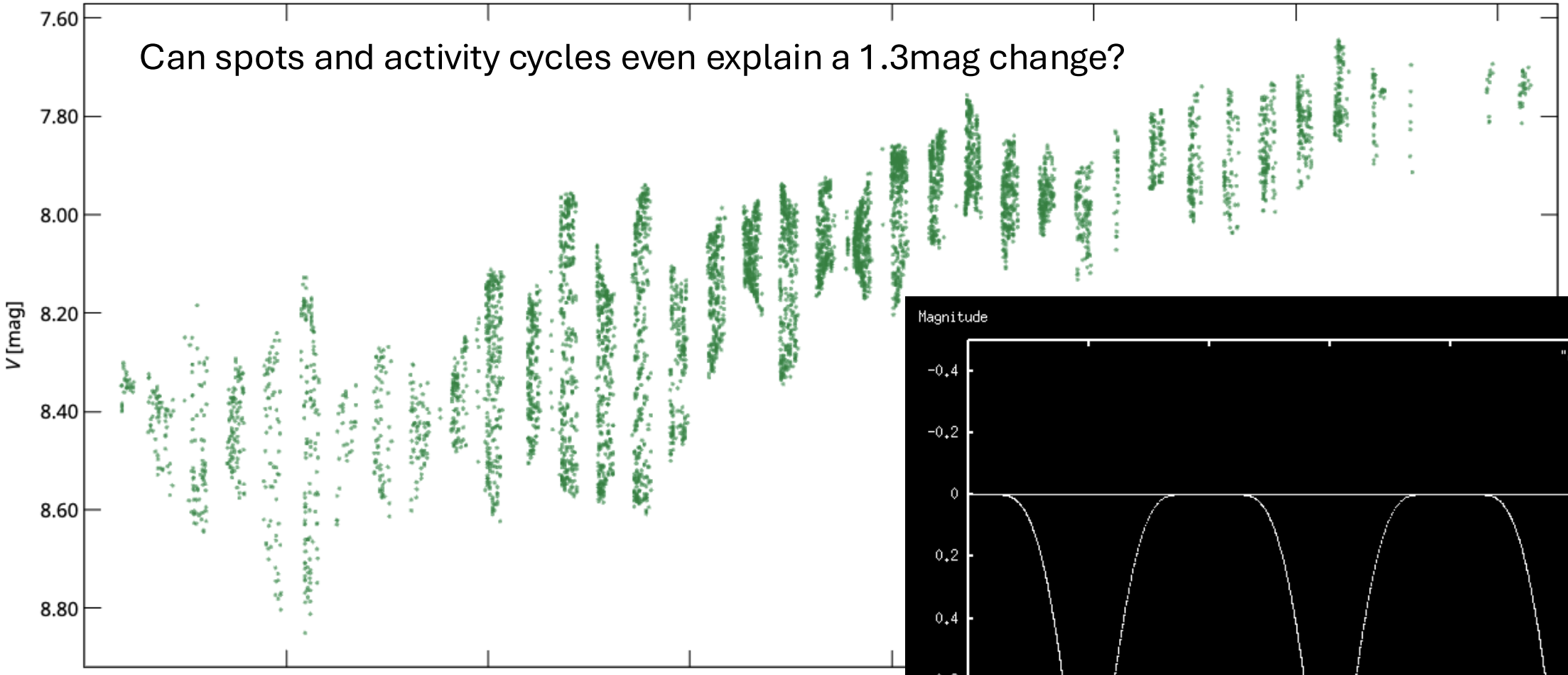
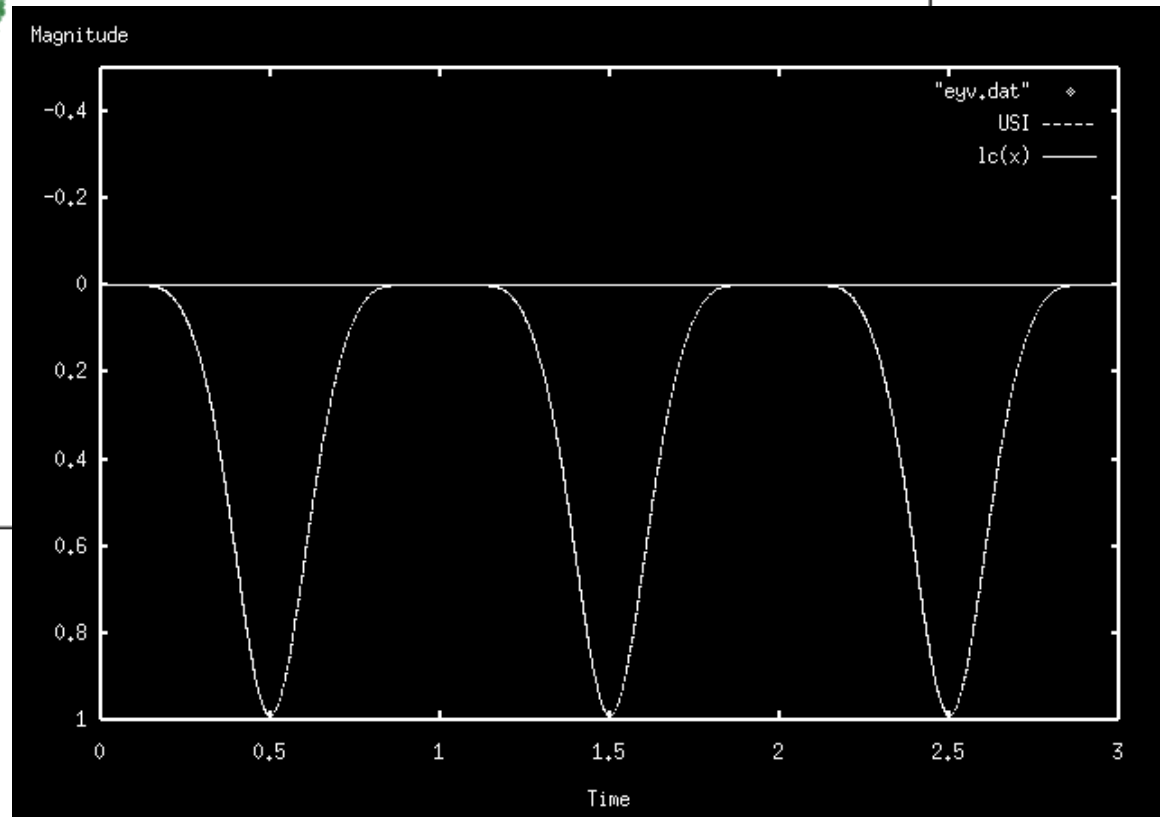


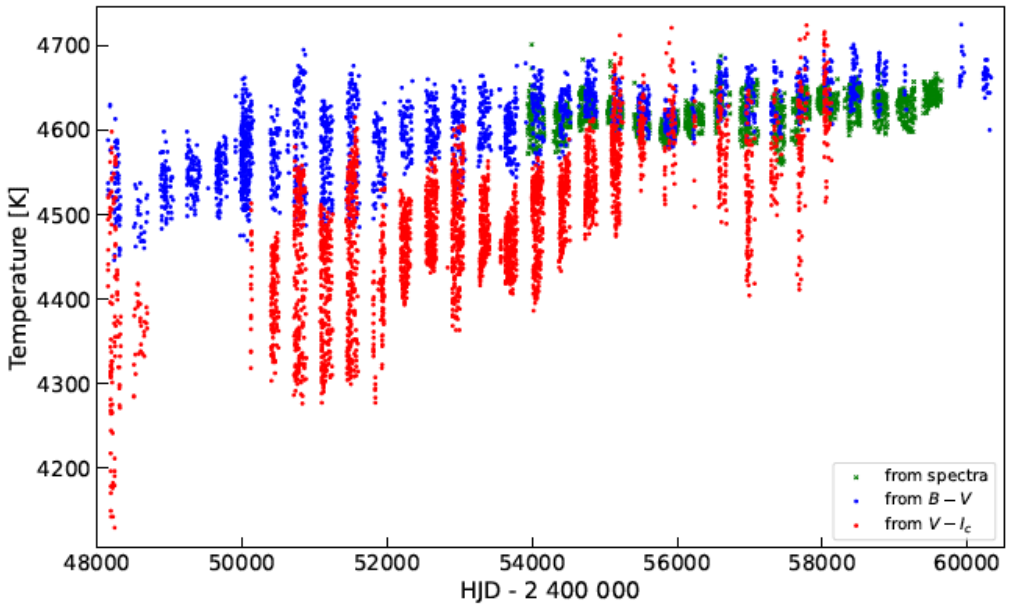


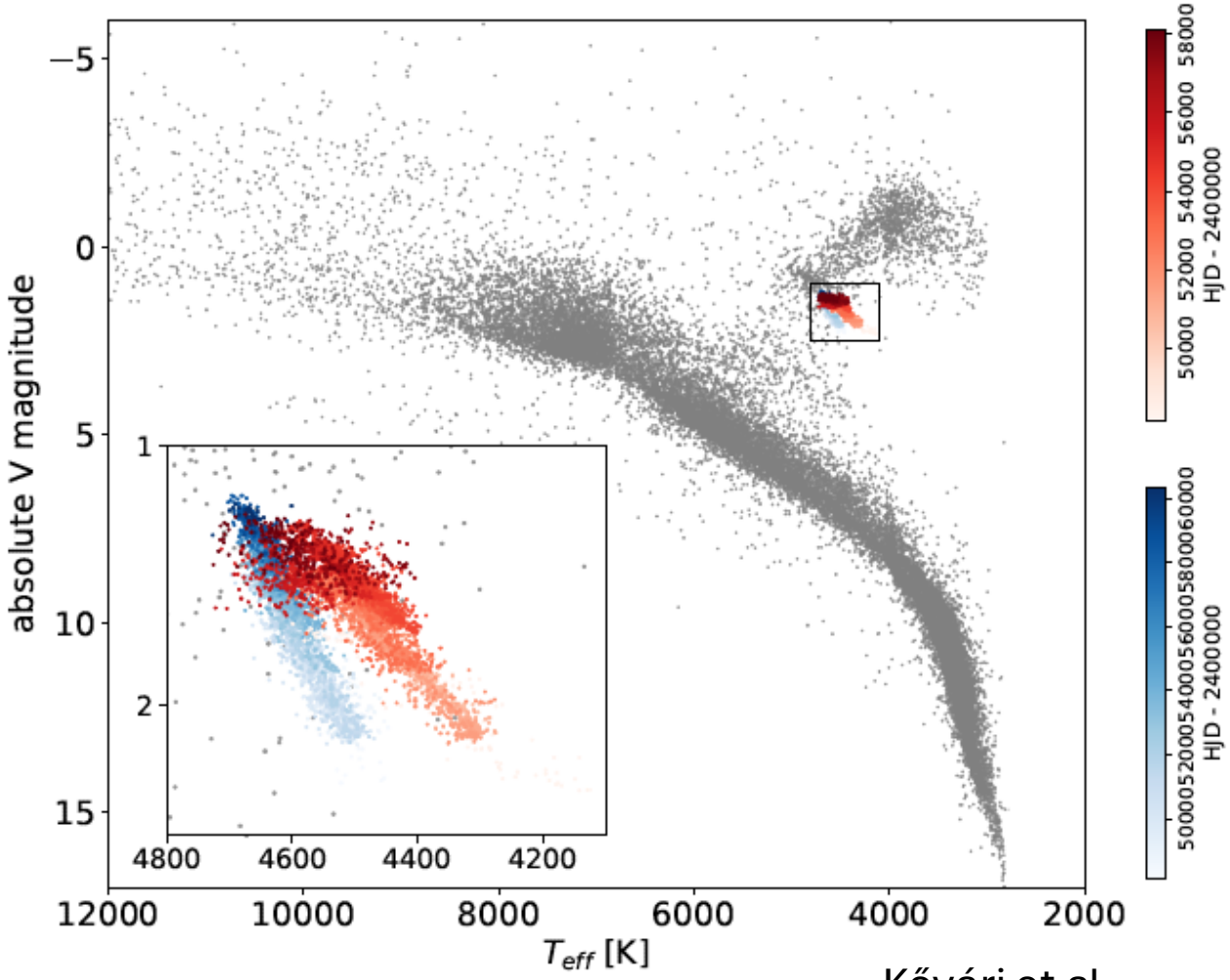
We can mix up a simple model with a pole-on star, but we need to cover half of it with Tsp=Tstar-2000K “black” spot – how realistic is that?

We should be careful when interpreting data – XX Tri (K0III SB)

The star moves noticably on the HRD during the observations

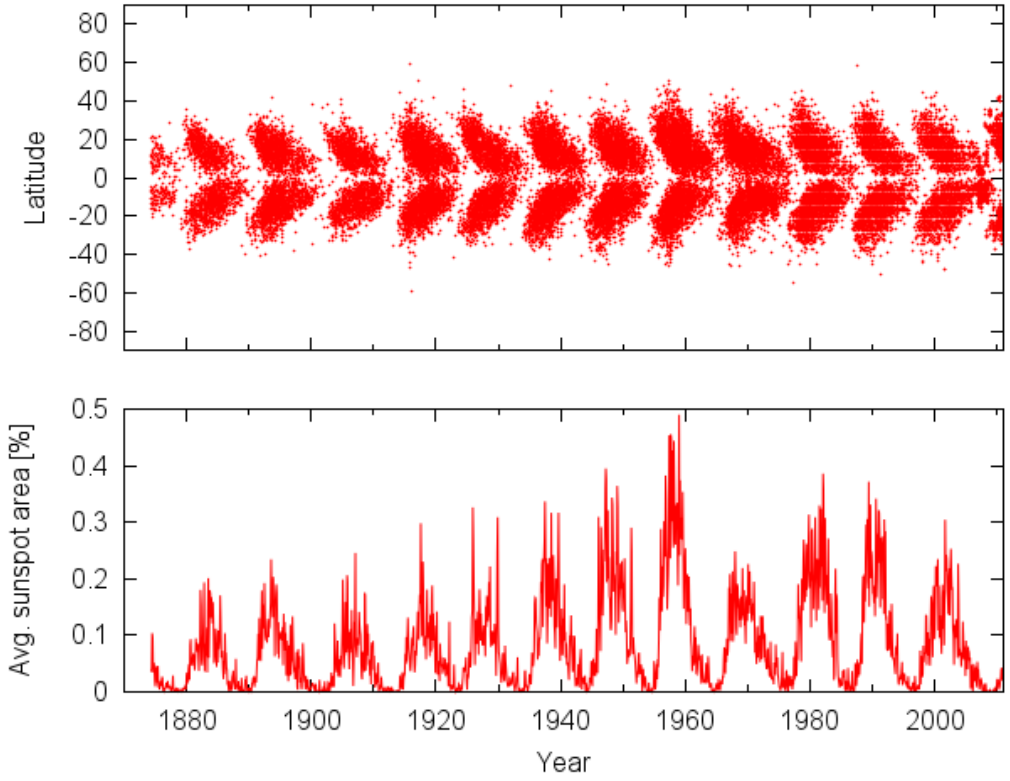
Constant unspotted brightness (used for models) could be misleading on long term



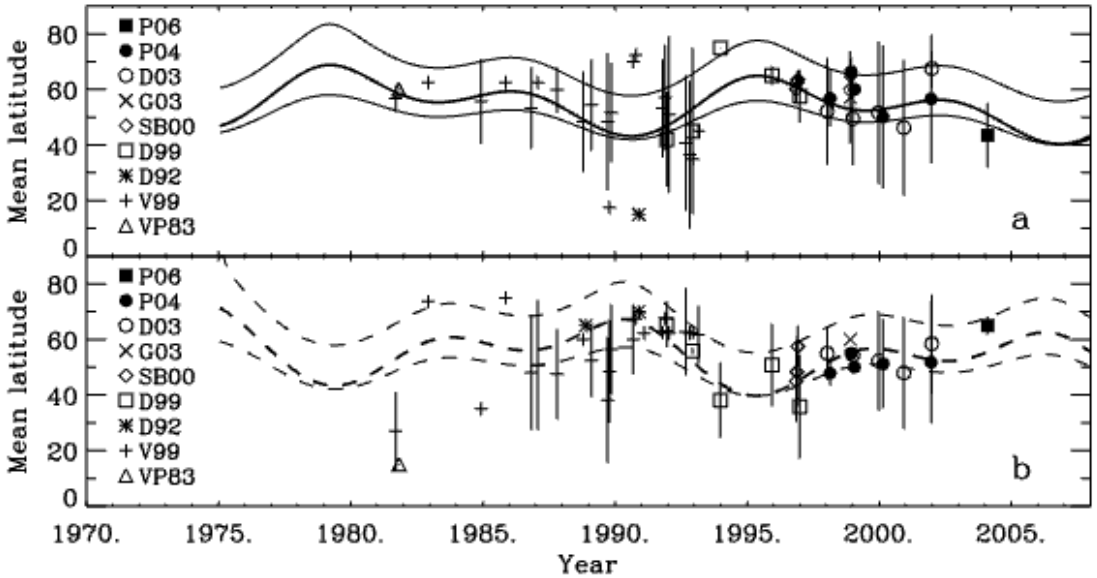


Kővári et al. 2025, submitted

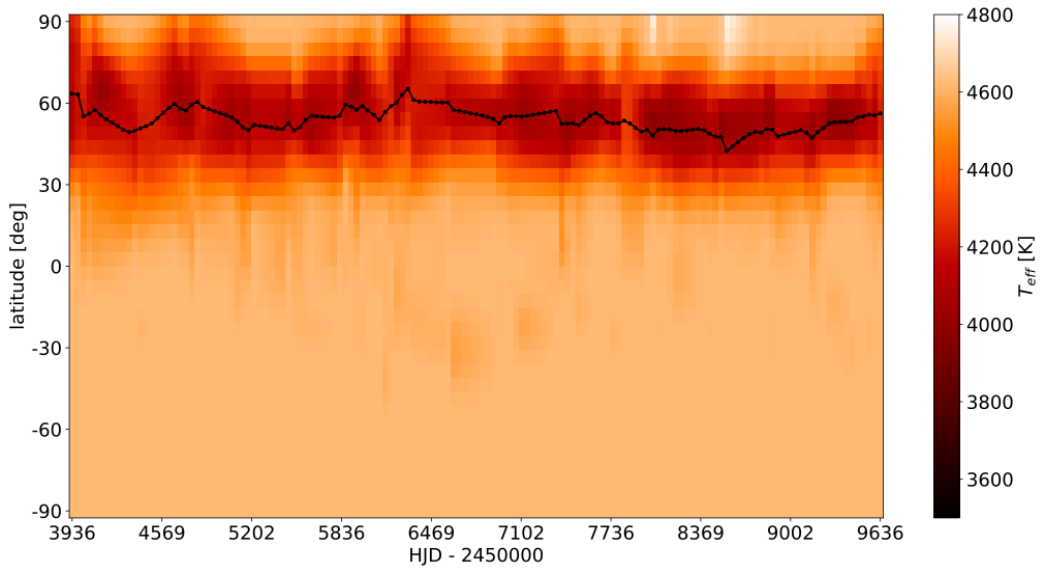
What about the butterfly diagram?



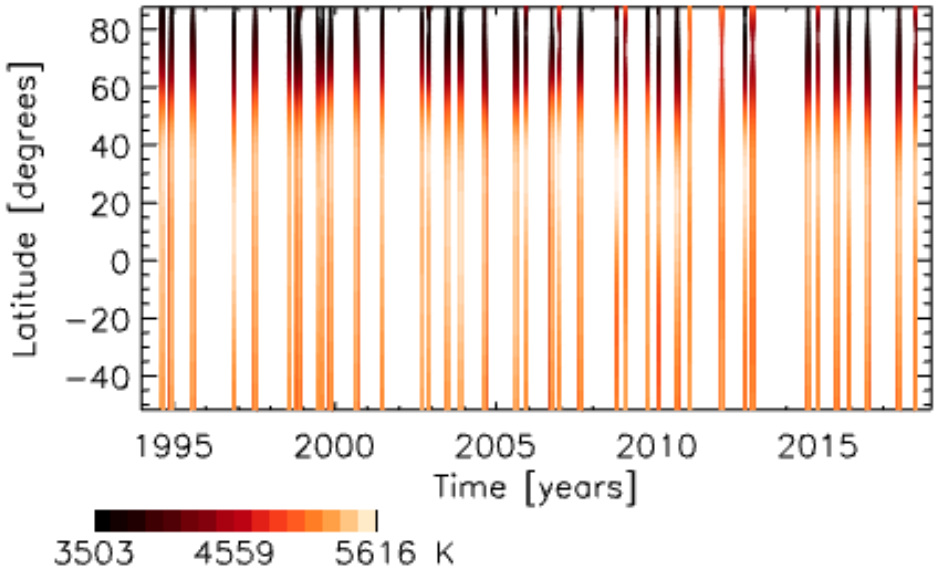
Photometric spot mapping provides limited information on latitudes, but Doppler-mapping can yield latitudes, basically giving us stellar butterfly diagrams



HR1099, Berdyugina et al. 2007



XX Tri, Kővári et al, in prep

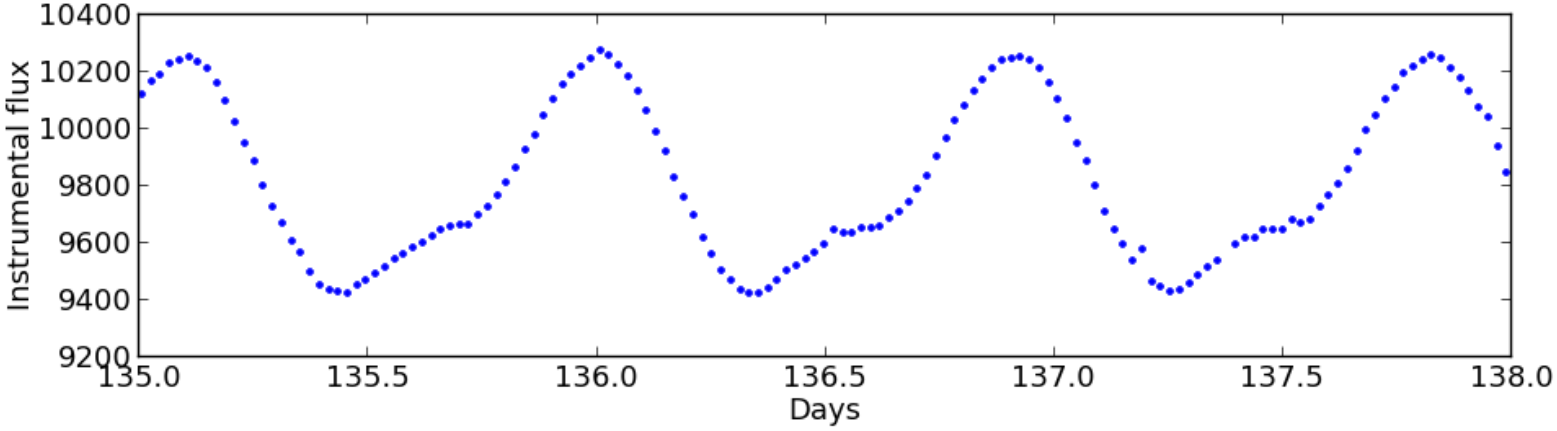


HD 199178, Hackman et al. 2019

Fig. 3. DIs averaged over longitudes vs. dates: Time-latitude diagram.

Can we use space photometry to detect cycles?

- More precise than ground-based data
- Has less rain/sunrise issues



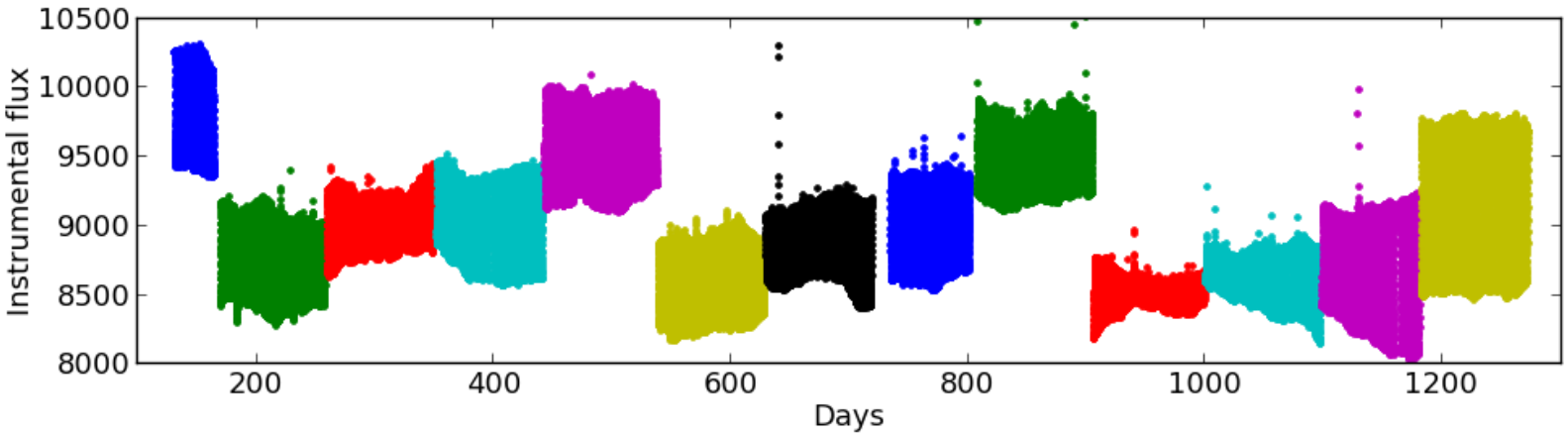
Can we use space photometry to detect cycles?

- More precise than ground-based data
- Has less rain/sunrise issues

Yes, but

- Has no filters (no temperature information)
- There are instrumental trends

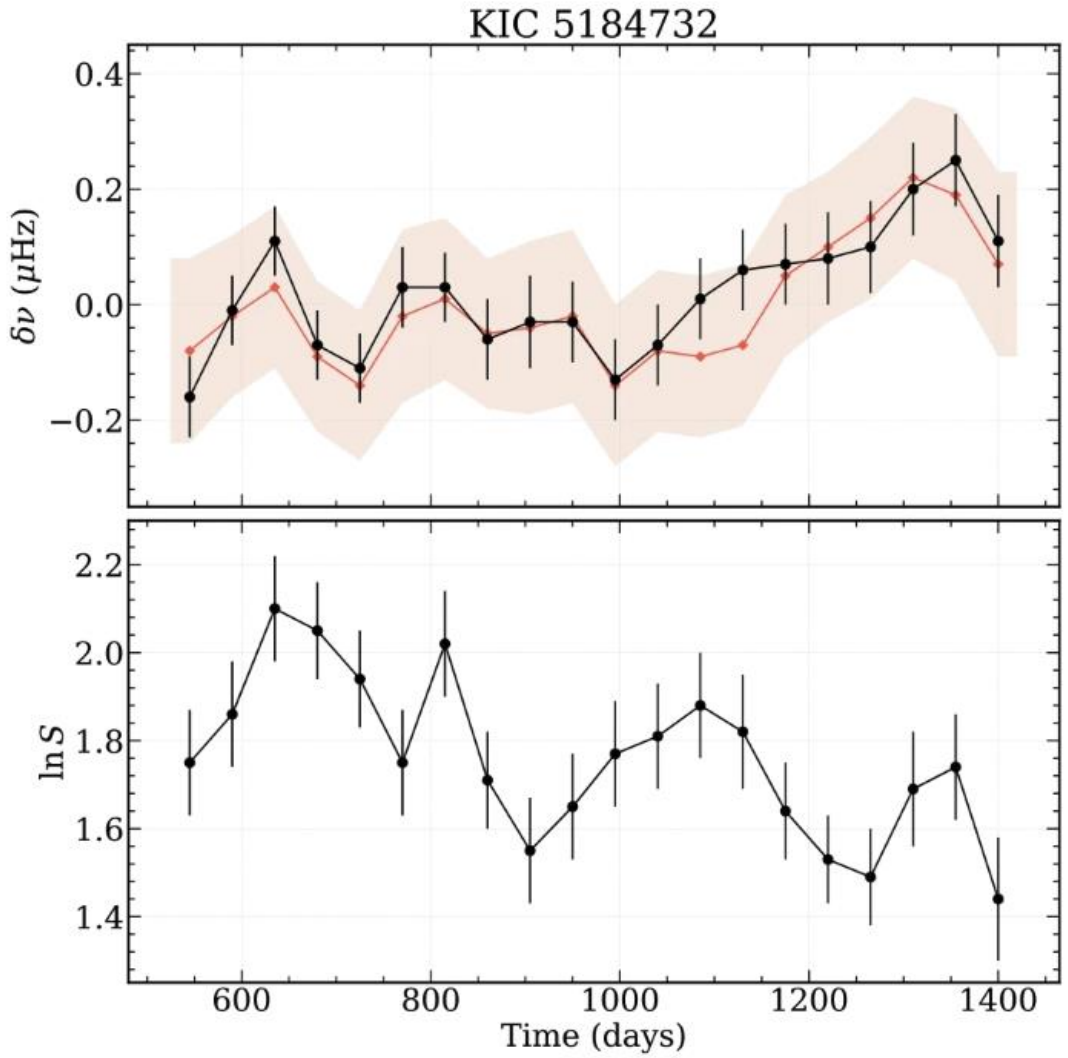
Can we still use them to find cycles?



Cycles on other stars – asteroseismology

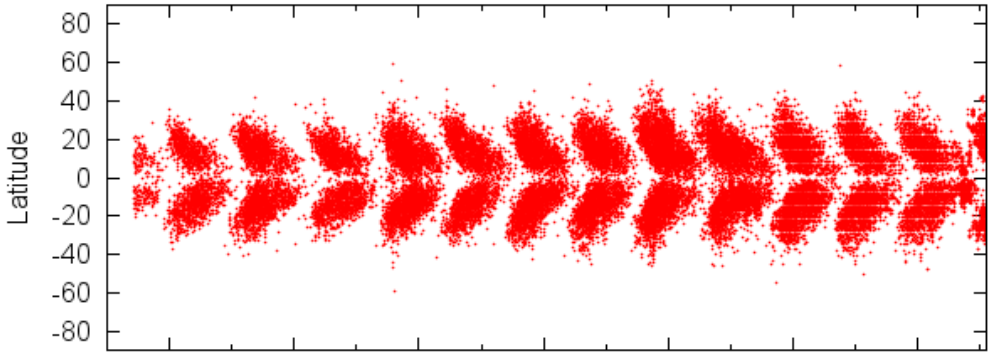
On solar-like (F-G) stars p-mode oscillations have been found to show a cyclic behavior

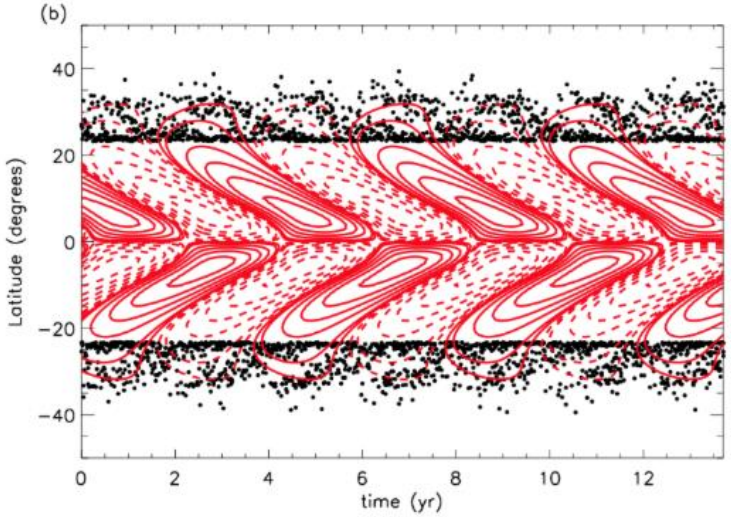
- oscillations driven by stochastic near-surface convective motions
- The frequencies are probably influenced by surface magnetic fields (spots?)



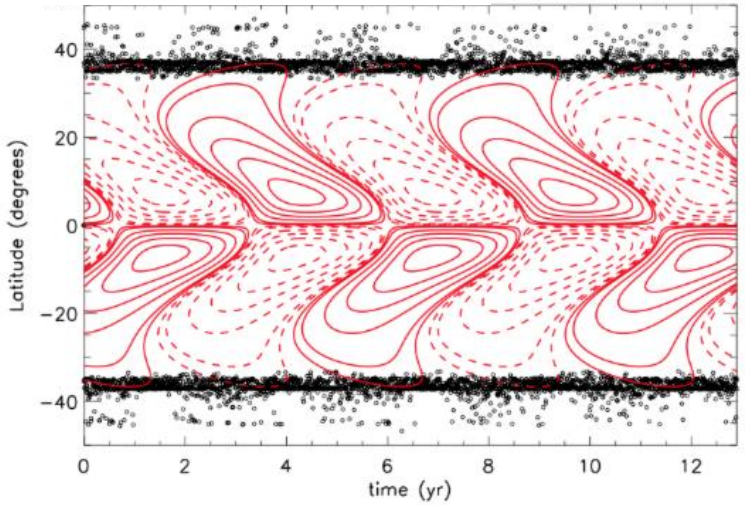
Jeffers et al. (2023); Santos et al. (2018)

Cycles on other stars – ”inverse” butterfly diagram





Fast-rotatig Sun (2d)

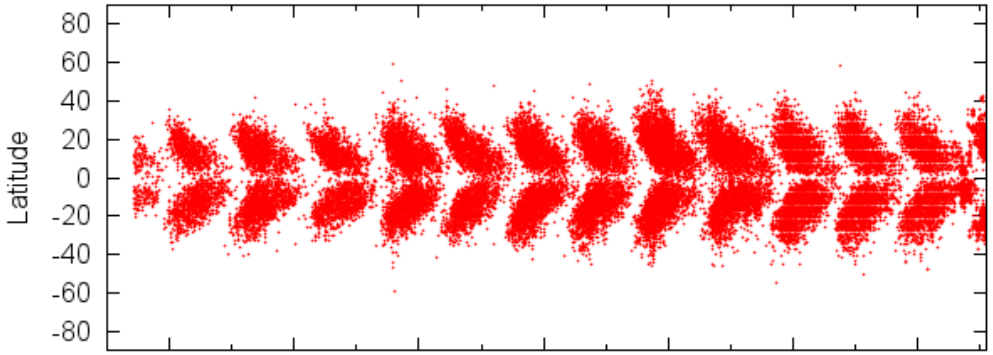


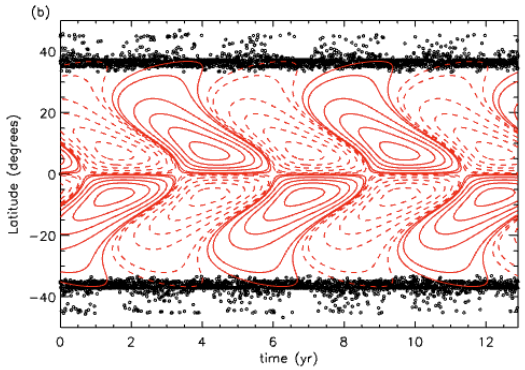
Fast-rotatig K0V star (2d)

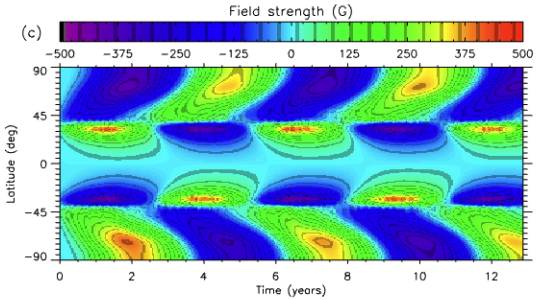
Işık,Schmitt,& Schüssler (2011)

On a fast-rotating late-type star the typical latitude could change in a way that can be detectable in the change of P_{rot} as a result of differential rotation

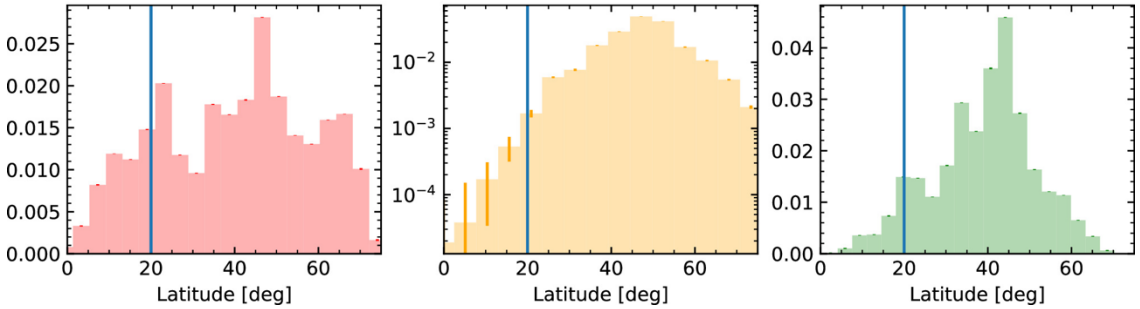
Cycles on other stars – ”inverse” butterfly diagram







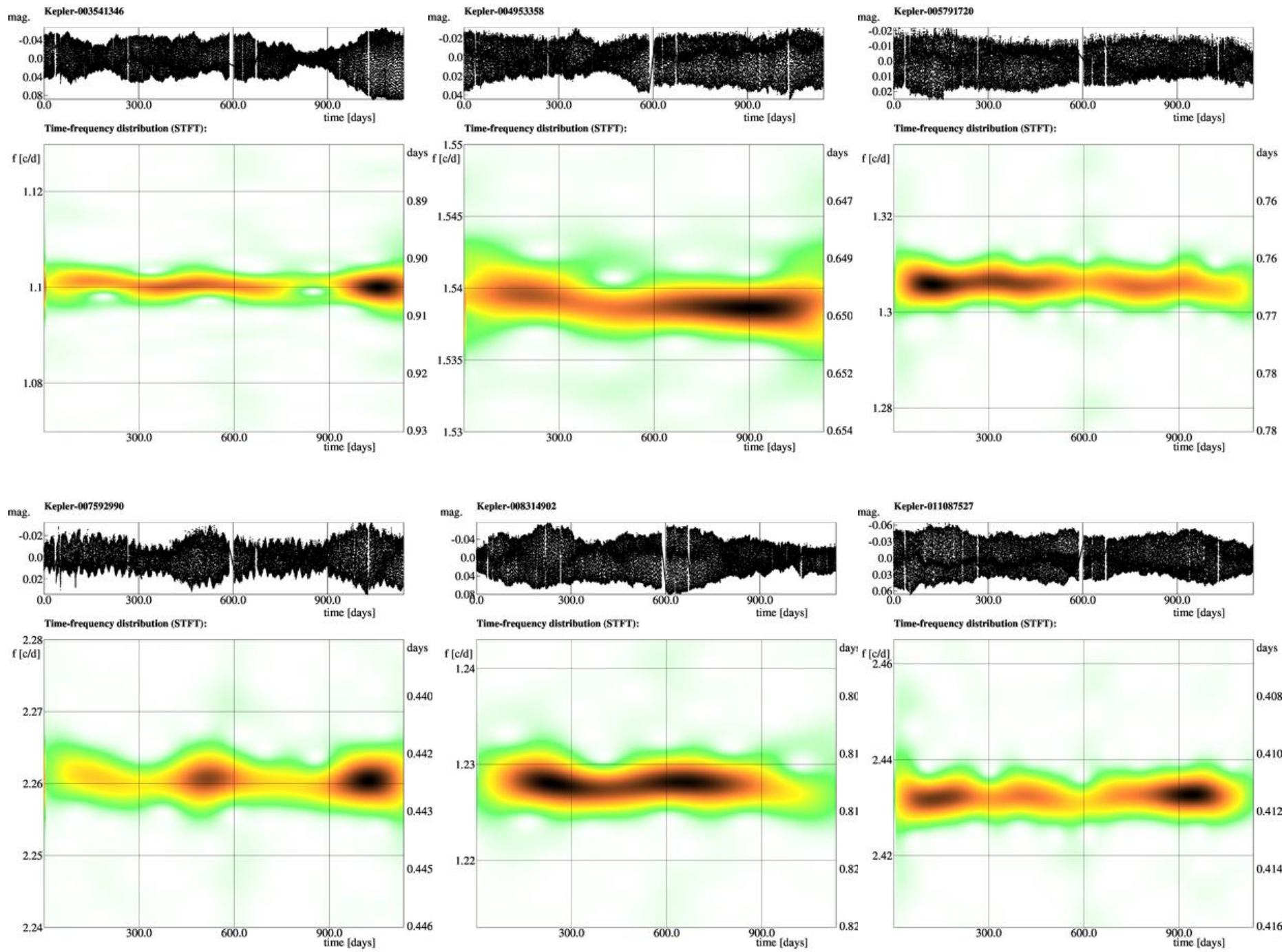
Emergence latitudes/radial surface magnetic field on a K0V star (Prot=3d) Işık,Schmitt,& Schüssler (2011)



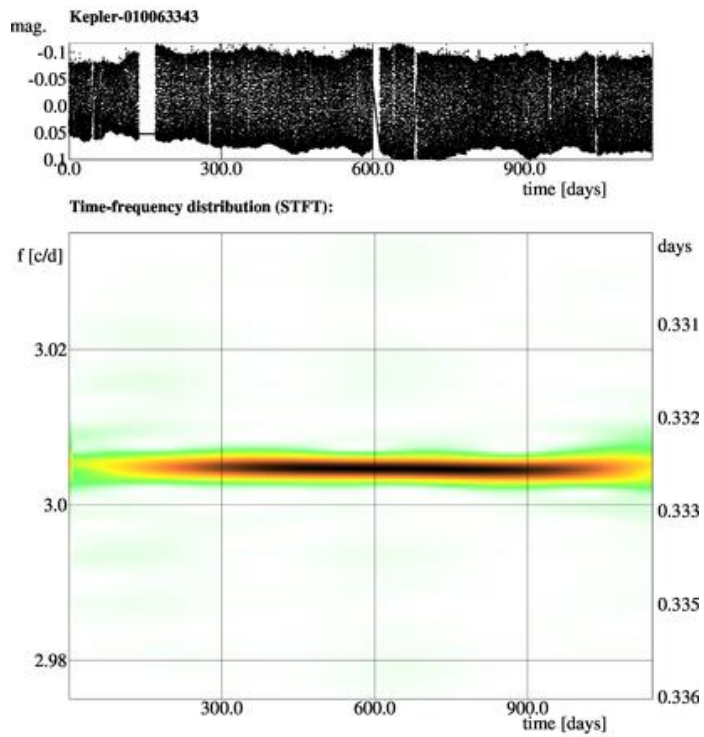
Possible emergence latitudes on a fast-rotating Sun (Prot=3d) Işık et al (2024)

On a fast-rotating late-type star the typical latitude could change in a way that can be detectable in the change of P_{rot} as a result of differential rotation

We found hints on cycles of 300–900 days on 9 fast-rotating Kepler-stars



Vida et al 2014



a comparison, where no change of Prot was seen

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- Unlike Kepler, continuous observation is rare
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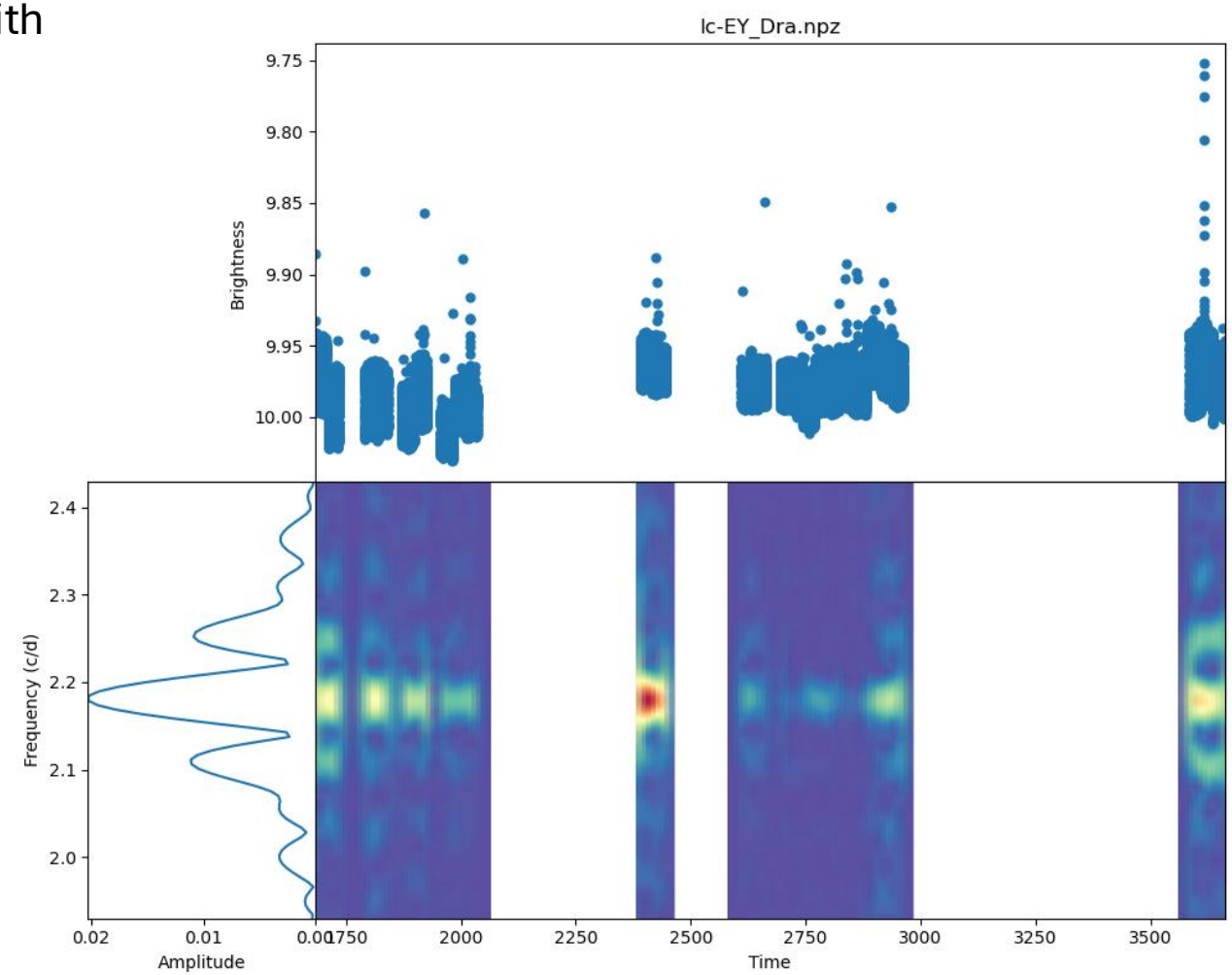

Vida, Seli et al in prep

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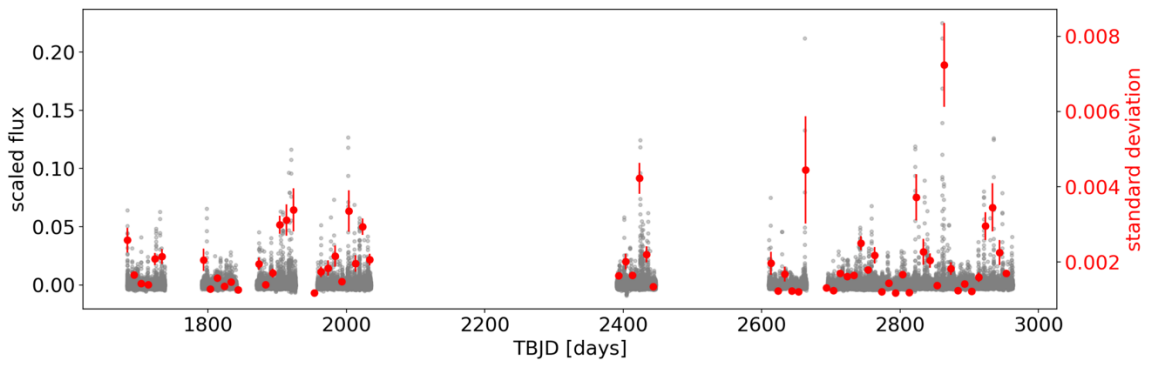

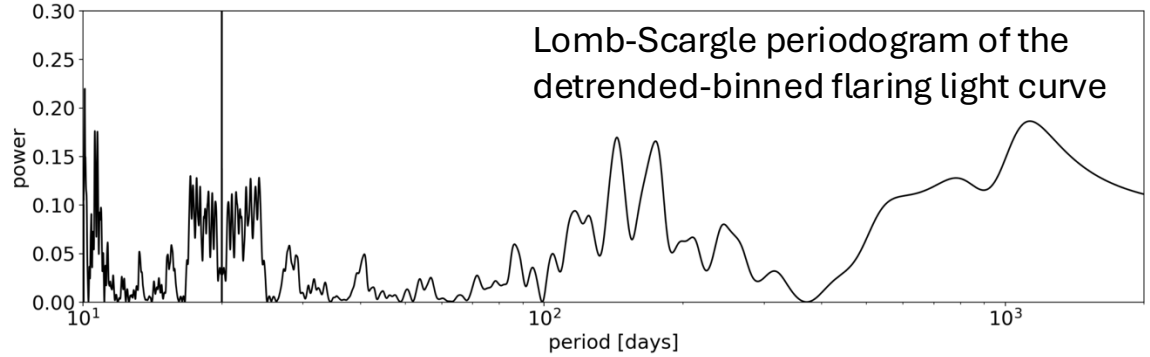

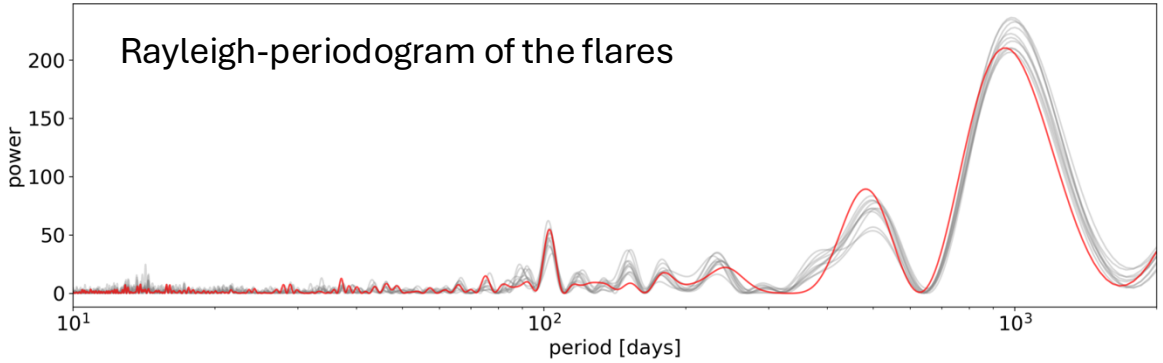

The 100-day period is likely an alias due to sampling

Vida, Seli et al in prep

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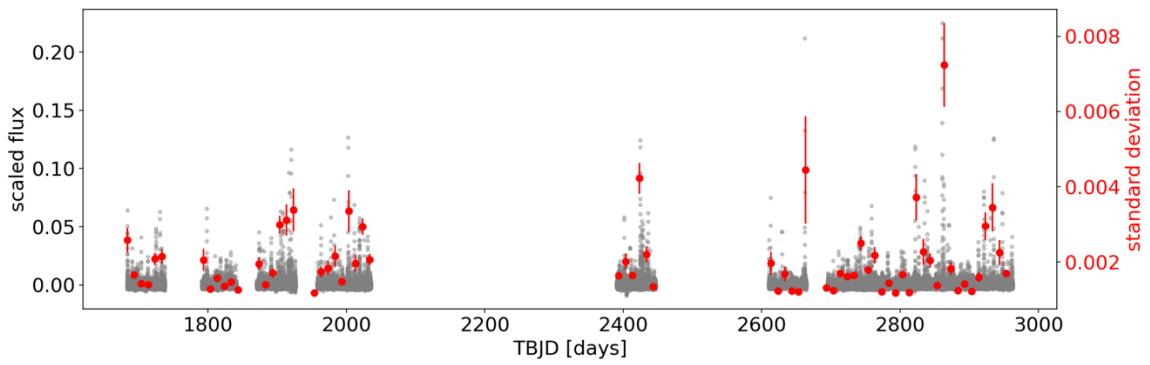
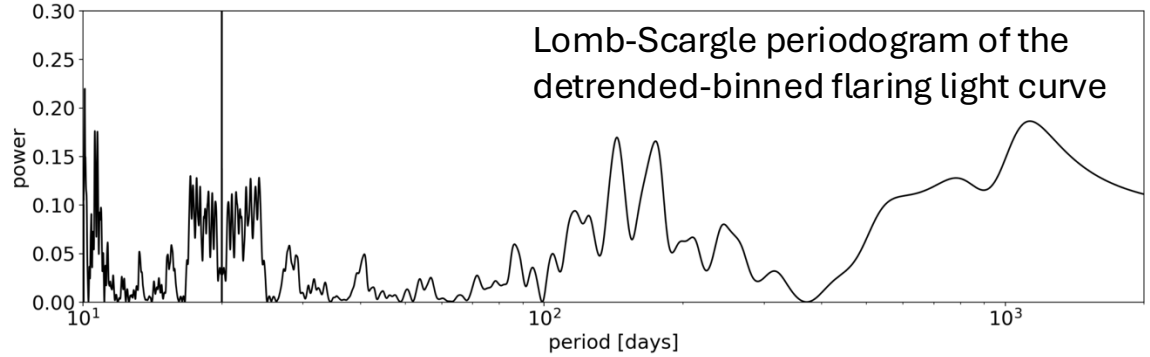
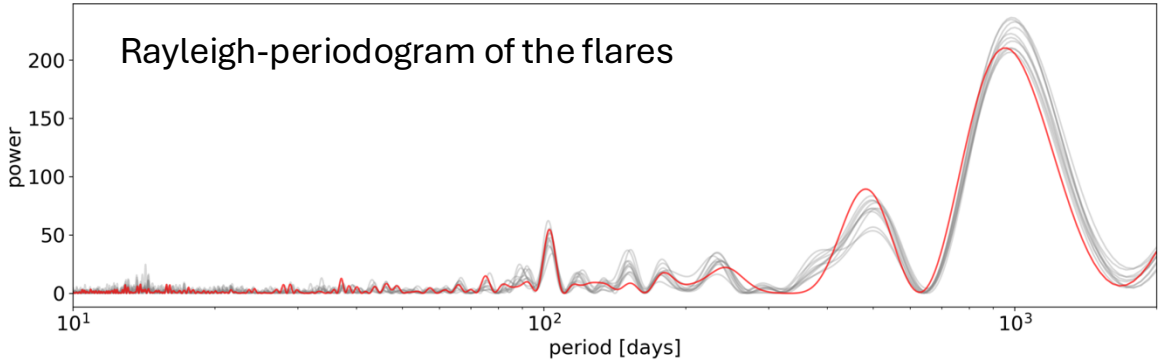
- Further ideas to try
  - Seismology? – the expected amplitude is in the μmag level, and the frequency range is too high to detect p-modes
  - Light curve amplitude variations?
  - Changing flare-frequency distributions?







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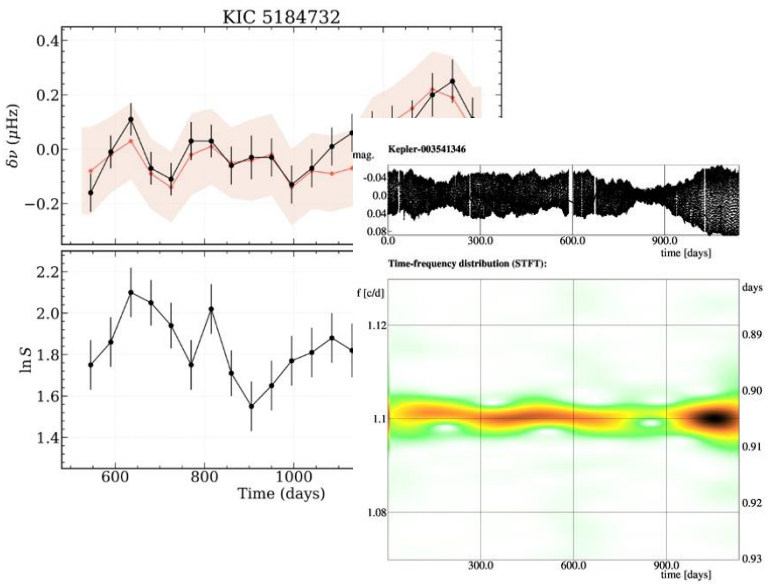
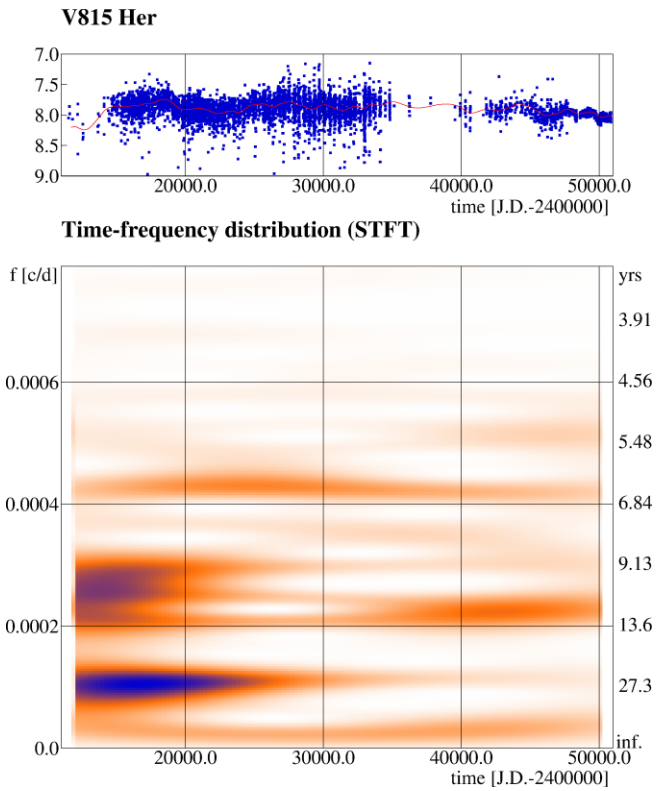
Takeaway message

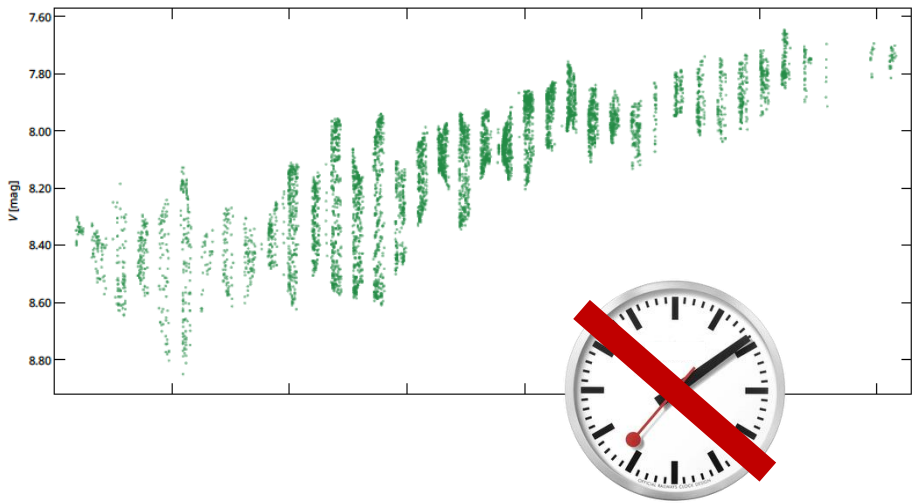
- Stellar activity cycles are important inputs for dynamo models
- Space photometry is useful in some cases, but time base is crucial – long-term, systematic observations are needed:
  - DASCH database
  - ASAS, APTs, OGLE…
  - Mt Wilson project



Cycles can be detected in numerous ways



Multiple, variable cycles can be seen in century-long data



Variable stars are not exact clockworks, we should be careful with quickly interpreting results